



Installation Instruction

Installation Instruction Number: 4908412

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QSK50 Tier 2 Dual Fuel Kit Installation Instructions

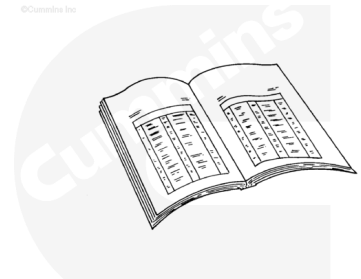
QSK50 Tier 2 Dual Fuel Kit Installation Instructions

General Information

The following procedure will provide instructions to assemble a Cummins® Hydraulic Fracturing Dual Fuel Kit onto an existing QSK50 Tier 2 base engine.

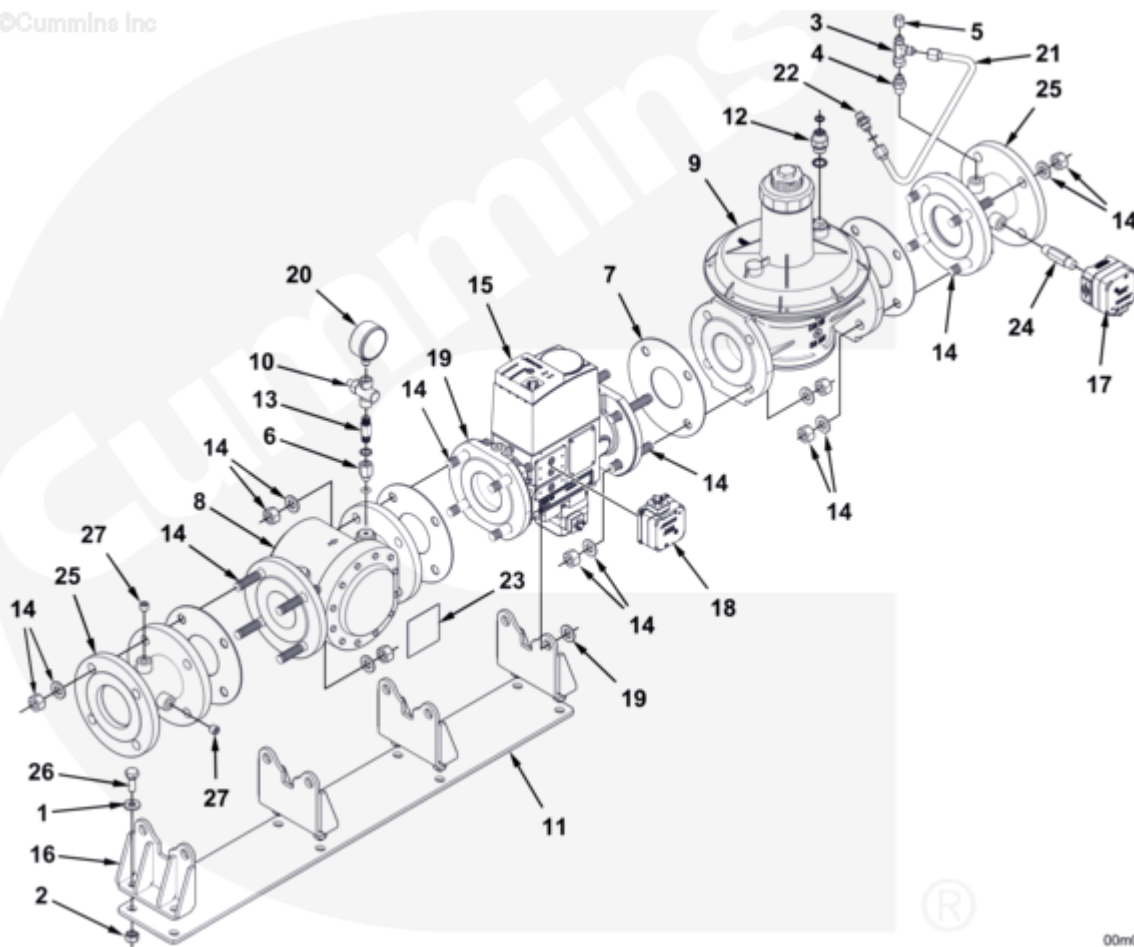
The QSK50 Tier 2 base engine should be in good working order prior to the installation of kit components. See the Dual Fuel Build Quality Sign-Off Sheet, Bulletin 4908413, in order to properly validate the working condition of the QSK50 Tier 2 base engine.

This procedure will only reference Cummins® factory-provided Dual Fuel Kit components. Additional parts may be necessary. They will be provided by the Cummins® distributor or the original equipment manufacturer (OEM) that is installing the kit. For all OEM and Cummins® distributor sourced components, reference available OEM and distributor documentation.



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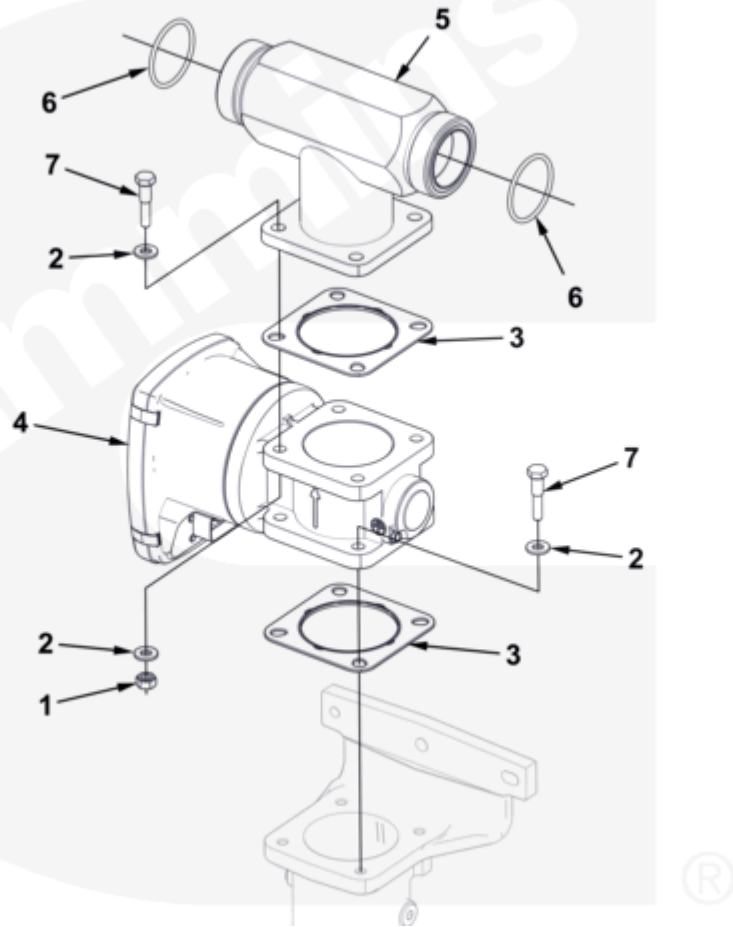
Component Diagrams



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Off-Engine Gas Train

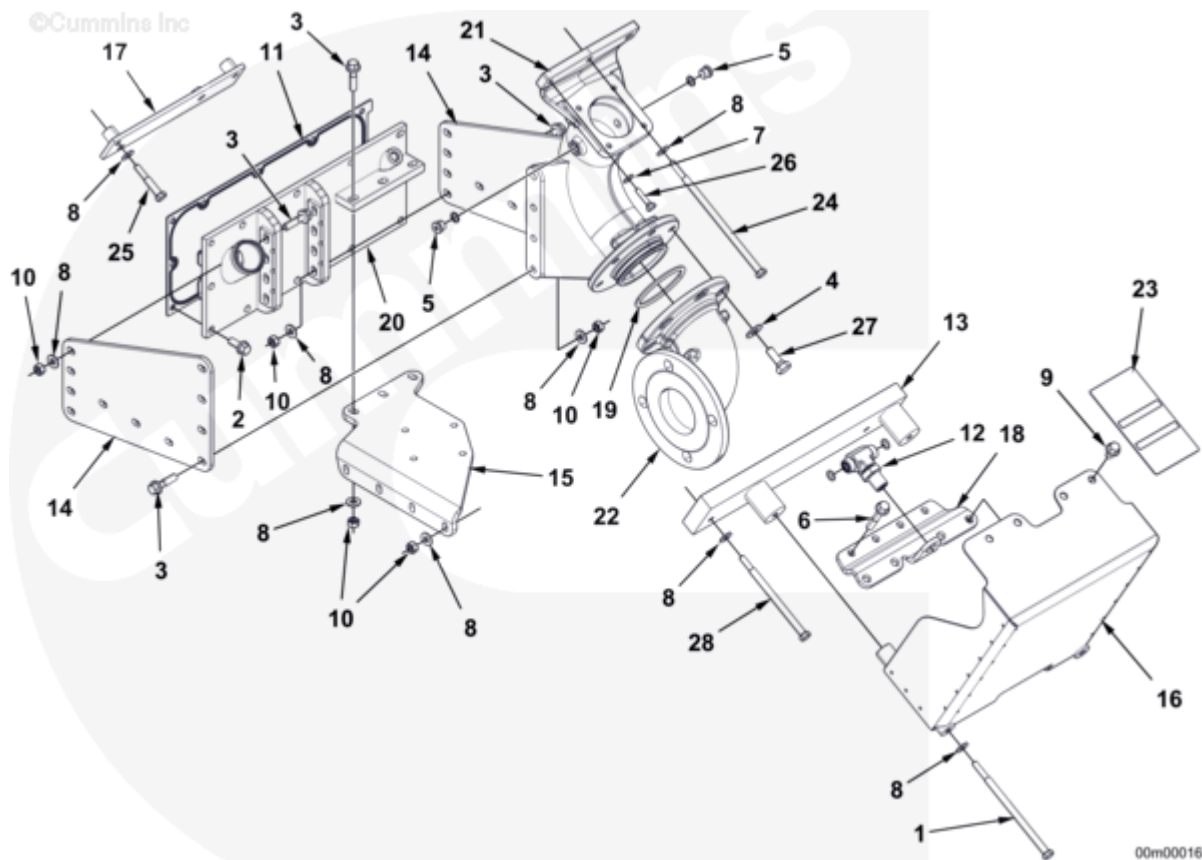
Reference Number	Item Number	Quantity	Item Name
1	3036900	8	Washer, Plain
2	3046208	8	Nut, Regular Hexagon
3	3063168	1	Tee, Female Run
4	3896807	1	Connector, Male
5	4000616	1	Cap, Tube
6	4322908	1	Adapter, Reducing Pipe
7	4322911	4	Gasket, Flange
8	4322912	1	Filter, Fuel
9	4322914	1	Valve, Pressure Regulator
10	4322915	1	Valve, Fuel Shutoff
11	4322947	1	Support, Accessory
12	4327782	1	Connector, Male
13	4327787	1	Nipple, Plain Hexagon
14	4327792	16	Assembly, Stud



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Fuel Control Valve

Reference Number	Item Number	Quantity	Item Name
1	69832	4	Nut, Regular Hexagon
2	3042328	12	Washer, Plain
3	3606534	2	Gasket, Connection
4	4322918	1	Governor, Woodward
5	4322970	1	Connection, Fuel Inlet
6	4327742	2	Seal, O-ring
7	S111	8	Screw, Hexagon Head Cap



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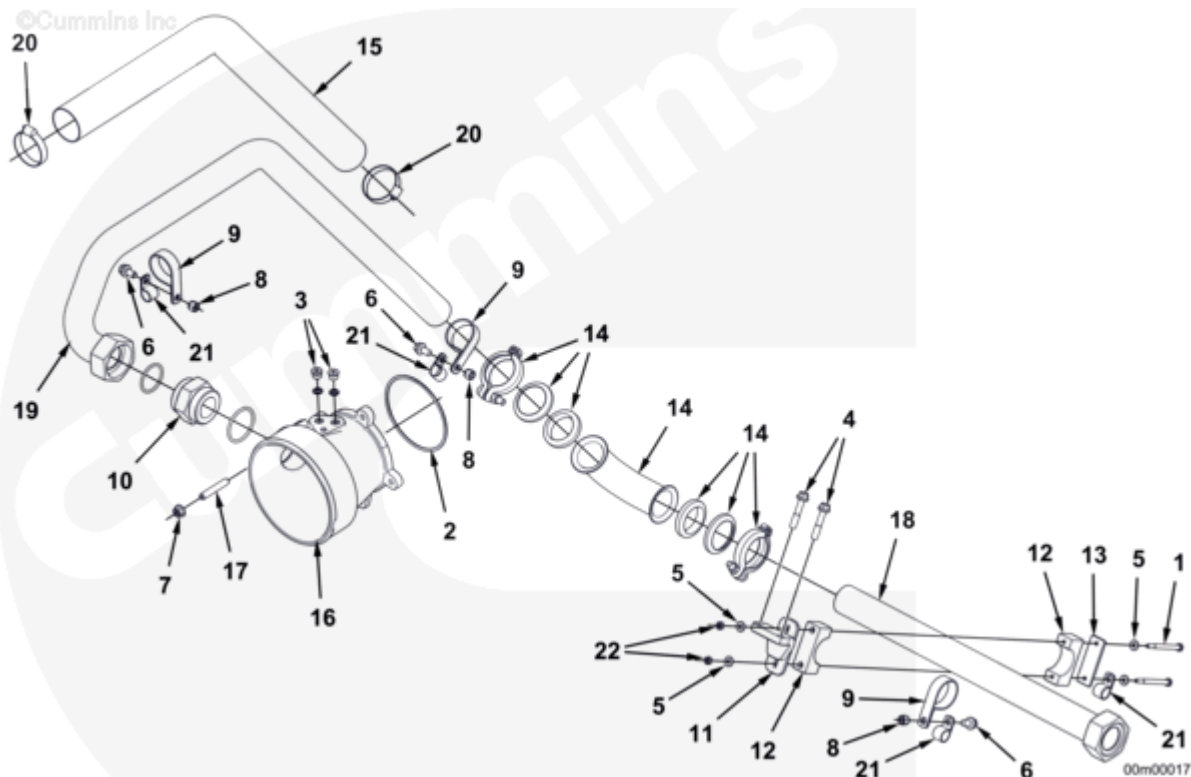
On-Engine Customer Connection

Reference Number	Item Number	Quantity	Item Name
1	137872	2	Screw, Hexagon Head Cap
2	3014766	12	Screw, Captive Washer Cap
3	3033038	22	Screw, Captive Washer Cap
4	3036900	6	Washer, Plain
5	3037536	2	Plug, Threaded
6	3040837	4	Screw, Captive Washer Cap
7	3042328	1	Washer, Plain
8	3042329	30	Washer, Plain
9	3069120	4	Screw, Captive Washer Cap
10	3201128	22	Nut, Regular Hexagon
11	3642347	1	Gasket, Hand Hole
12	3042328	1	Tee, Male Union
13	4327795	1	Bracket, Debris Shield
14	4327826	2	Brace, Support

On-Engine Customer Connection			
Reference Number	Item Number	Quantity	Item Name
15	4327827	1	Brace, Support
16	4327828	1	Shield, Debris
17	4327832	1	Bracket, Breather
18	4328662	1	Bracket, Debris Shield
19	4328686	1	Seal, O-ring
20	4334843	1	Cover, Hand Hole
21	4334859	1	Connection, Fuel Transfer
22	4334861	1	Connection, Fuel Transfer
23	4350851	1	Dataplate
24	4968985	2	Screw, Hexagon Head Cap
25	S104	2	Screw, Hexagon Head Cap
26	S119	1	Screw, Hexagon Head Cap
27	S145	6	Screw, Hexagon Head Cap
28	S170A	2	Screw, Hexagon Head Cap

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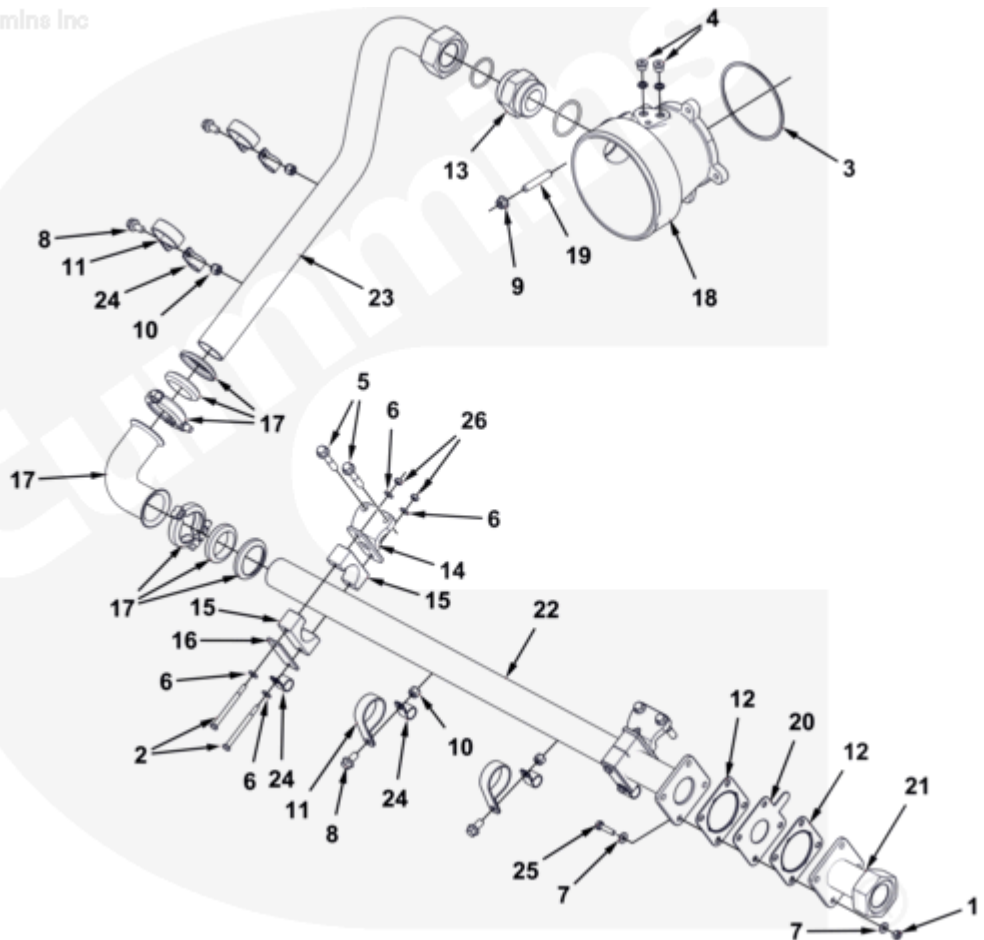
Left Bank Gas Inlet Tubes

Reference Number	Item Number	Quantity	Item Name
1	137719	2	Screw, Hexagon Head Cap
2	3029640	1	Seal, O-ring
3	3037536	2	Plug, Threaded
4	3040837	2	Screw, Captive Washer Cap
5	3042327	4	Washer, Plain
6	3064321	3	Screw, Captive Washer Cap
7	3063828	1	Nut, Hexagon Flange
8	3175239	3	Nut, Lock
9	3202124	3	Clip
10	4322974	1	Connector, Male
11	4322977	1	Brace, Tube
12	4322978	1	Clamp, Tube
13	4322979	1	Plate, Clamping
14	4327732	1	Elbow, Tube Connector
14	4327754	1	Shield, Heat
16	4327788	1	Connection, Air Intake

Left Bank Gas Inlet Tubes

Reference Number	Item Number	Quantity	Item Name
17	4328541	1	Stud, Continuous Thread
18	4350823	1	Tube, Gas Inlet
19	4350825	1	Tube, Gas Inlet
20	43828D	2	Clamp, Hose
21	69911A	4	Clip
22	S270	2	Nut, Regular Hexagon
22	4334861	1	Connection, Fuel Transfer

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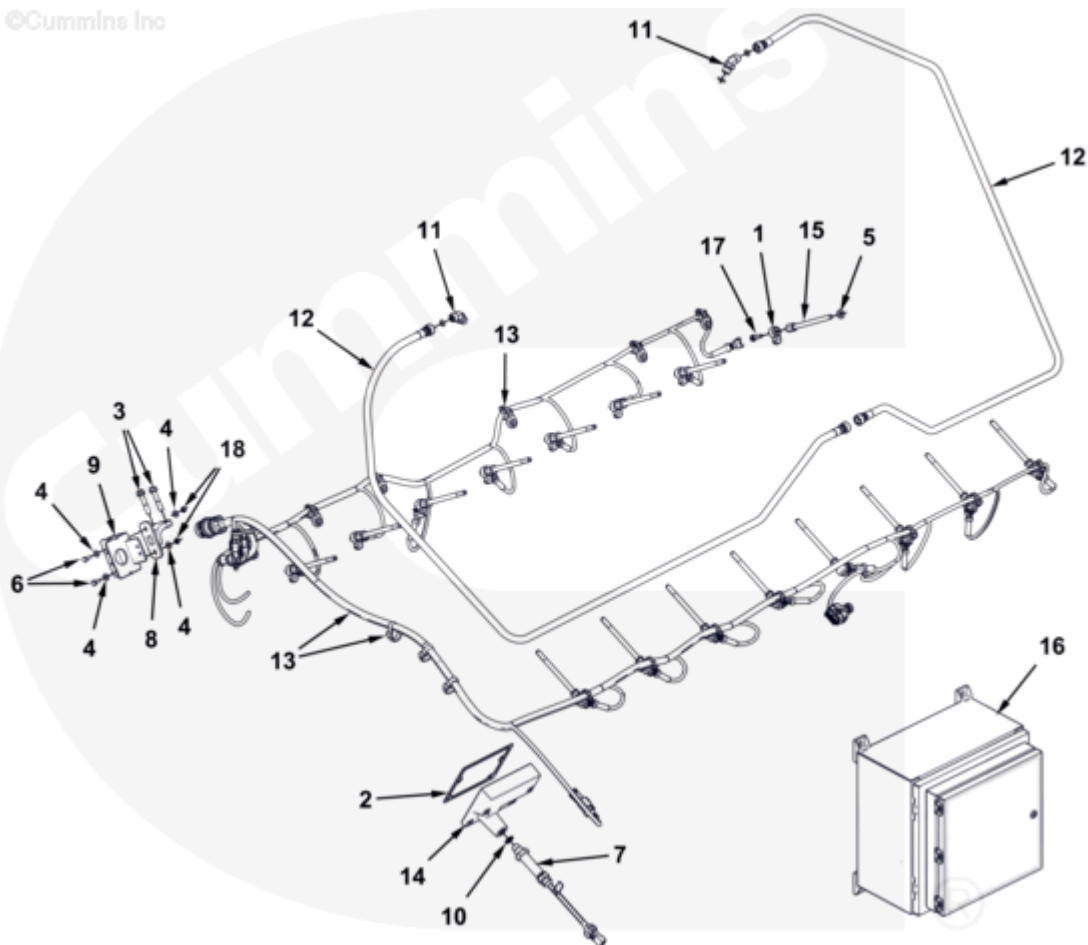


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Right Bank Gas Inlet Tubes

Reference Number	Item Number	Quantity	Item Name
1	69832	4	Nut, Regular Hexagon

Right Bank Gas Inlet Tubes			
Reference Number	Item Number	Quantity	Item Name
2	137719	4	Screw, Hexagon Head Cap
3	3029640	1	Seal, O-ring
4	3037536	2	Plug, Threaded
5	3040837	4	Screw, Captive Washer Cap
6	3042327	8	Washer, Plain
7	3042328	1	Washer, Plain
8	3064321	4	Screw, Captive Washer Cap
9	3093828	1	Nut, Hexagon Flange
10	3175239	4	Nut, Lock
11	3202124	4	Clip
12	3606534	2	Gasket, Connection
13	43229974	1	Connector, Male
14	4322977	2	Brace, Tube
15	4322978	2	Clamp, Tube
16	4322979	2	Plate, Clamping
17	4327732	1	Elbow, Tube Connector
18	4327788	1	Connection, Air Intake
19	4328541	1	Stud, Continuous Thread
20	4328544	1	Orifice, Fuel Flow Limiting
21	4335202	1	Tube, Gas Inlet
22	4350830	1	Tube, Gas Inlet
23	4350832	1	Tube, Gas Inlet
24	69911A	6	Clip
25	S119	4	Screw, Hexagon Head Cap
26	S270	4	Nut, Regular Hexagon



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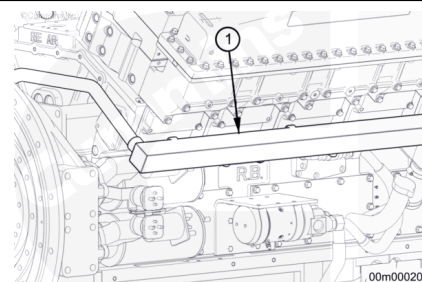
On-Engine Wiring Harness

Reference Number	Item Number	Quantity	Item Name
1	2897654	16	Sensor, Knock
2	3040721	1	Gasket, Cam Follower Housing
3	3040837	2	Screw, Captive Washer Cap
4	3042327	4	Washer, Plain
5	4326355	1	Seal, O-ring
6	4087931	2	Screw, Hexagon Head Cap
7	4307676	1	Sensor, Position
8	4322977	1	Brace, Tube
9	4322996	1	Bracket, Wiring Retainer
10	4326355	1	Seal, O-ring
11	4327748	2	Elbow, Male Union
12	4327802	2	Hose, Flexible
13	4334865	1	Harness, Wiring
14	4350835	1	Cover, Camshaft

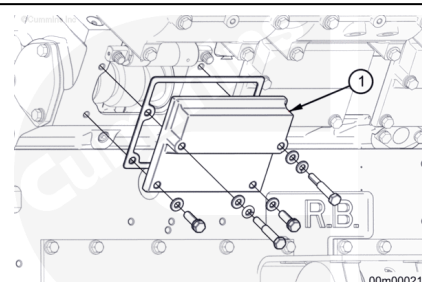
On-Engine Wiring Harness			
Reference Number	Item Number	Quantity	Item Name
15	S119	16	Screw, Hexagon Head Cap
16	4358036	1	Panel, Control
17	4358010	16	Screw, Hexagon Head Cap
18	S270	2	Nut, Regular Hexagon

Cylinder 16 Camshaft Cover and Sensor

Remove the right bank engine wire harness (1) to allow access to the cylinder 16 camshaft cover. Refer to Procedure 019-043 in Section 19 (/qs3/pubsys2/xml/en/procedures/330/330-019-043.html) in the QSK50 Service Manual. Retain the mounting capscrews.



Remove the original QSK50 cylinder 16 (R8) camshaft cover (1). Discard the gasket. Keep the original cover capscrews.

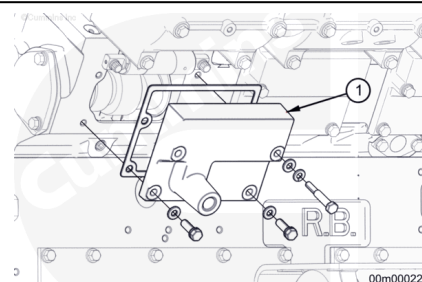


Install the Dual Fuel cylinder 16 camshaft cover (1) using a new camshaft cover gasket.

Install the original three short camshaft cover capscrews.

Torque Value: 40 n·m [30 ft-lb]

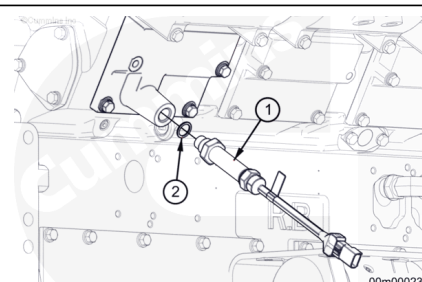
Note : The top left camshaft cover bolt hole will be used when the engine wire harness is installed.



Install the Dual Fuel camshaft position sensor (1) and o-ring, Cummins® Part Number 4326355 (2). Thread the sensor into the camshaft cover sensor bung until seated.

Torque Value: 73 n·m [54 ft-lb]

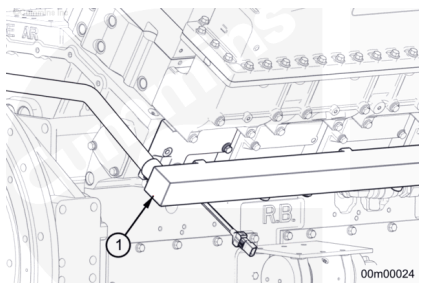
Note : The Dual Fuel camshaft position sensor comes with a pre-assembled jam nut. The jam nut is not used



for sensor installation. Discard the jam nut.

Install the QSK50 right bank engine harness (1). Make sure the capscrew is installed into the top left camshaft cover bolt hole. Refer to Procedure 019-043 in Section 19 (/qs3/pubsys2/xml/en/procedures/255/255-019-043-tr.html) in the QSK50 Service Manual.

Torque Value: 40 n•m [30 ft-lb]



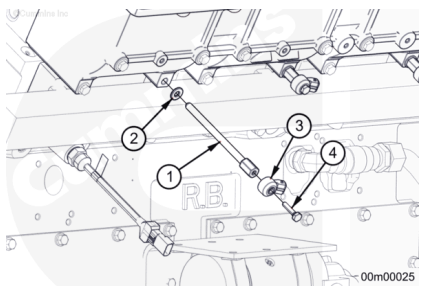
Knock Sensors

Remove the existing QSK50 forward intake manifold bolts and washer from each cylinder, 16 total.

Install the 16 hex head aftercooler capscrews (1). Make sure washer (2), Cummins® Part Number 3042329, is placed between the capscrew and the aftercooler.

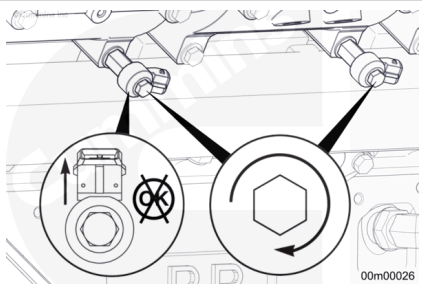
Torque Value: 45 n•m [33 ft-lb]

Install the knock sensor (3) and mounting capscrew, Cummins® Part Number S119 (4). Leave hand-tight.



Each of the 16 knock sensors **must** be clocked in a specific direction.

Knock Sensor Clocking		
Bank Cylinder Number	Knock Connector	Knock Sensor Clocking
Left Bank Cylinder 1	C22-1	4 o'clock
Left Bank Cylinder 2	C22-3	8 o'clock
Left Bank Cylinder 3	C22-5	9 o'clock
Left Bank Cylinder 4	C22-7	8 o'clock
Left Bank Cylinder 5	C22-9	8 o'clock
Left Bank Cylinder 6	C22-11	8 o'clock
Left Bank Cylinder 7	C22-13	8 o'clock



Knock Sensor Clocking		
Bank Cylinder Number	Knock Connector	Knock Sensor Clocking
Left Bank Cylinder 8	C22-15	8 o'clock
Right Bank Cylinder 1	C22-1	4 o'clock
Right Bank Cylinder 2	C22-1	4 o'clock
Right Bank Cylinder 3	C22-1	4 o'clock
Right Bank Cylinder 4	C22-1	4 o'clock
Right Bank Cylinder 5	C22-1	3 o'clock
Right Bank Cylinder 6	C22-1	8 o'clock
Right Bank Cylinder 7	C22-1	4 o'clock
Right Bank Cylinder 8	C22-1	4 o'clock

Tighten each knock sensor.

Torque Value: 20 n·m [177 in-lb]

Note : Do not orient the connector upward between 9 o'clock and 3 o'clock, Failure to follow this instruction may result in sensor water intrusion.

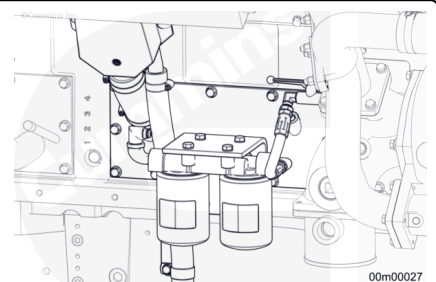
On-Engine Customer Connection

General Information

The hand hole cover, corrosion filters, and crankcase breather will need to be moved and/or modified to meet the Dual Fuel installation requirements.

All brackets and components are provided for full on-engine customer connection assembly.

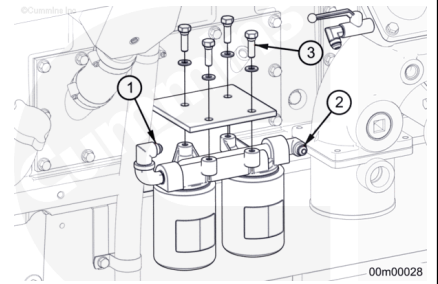
The illustration shows the QSK50 base engine breather, hand hole cover, and corrosion filters.



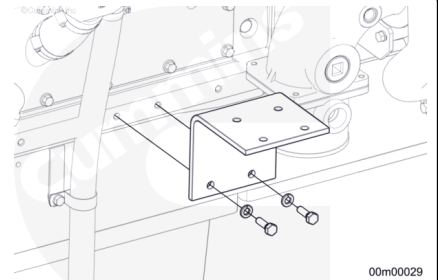
Close the ball valves of the corrosion filter supply and return lines, at the water pump.

Remove the fittings at the filter head supply (1) and return (2).

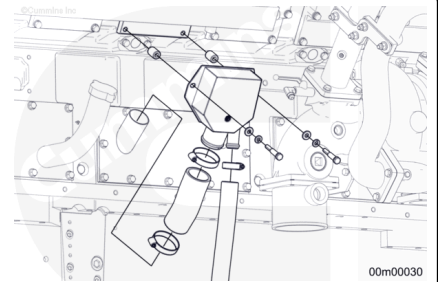
Remove the corrosion filter assembly by loosening the four capscrews (3) at the filter head and water supply. Keep these capscrews for reuse.



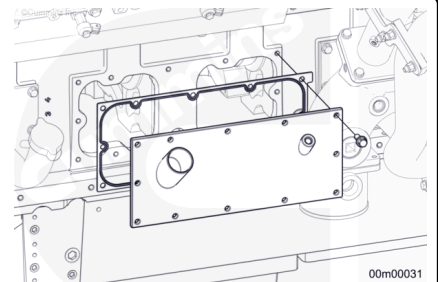
Remove and discard the corrosion resistor L-bracket mounting plate. A new mounting plate and location will be provided.



Remove the crankcase breather and breather hoses. The breather hoses will be reused in this installation. Keep the crankcase breather capscrews for reuse. New capscrews for the R4 and R5 camshaft covers are provided.



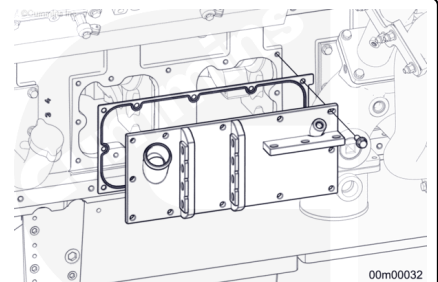
Remove and discard the QSK50 base engine hand hole cover and gasket. A new hand hole cover specific to the Dual Fuel System is provided.



Install the Dual Fuel hand hole cover and a new hand hole cover gasket onto the engine block.

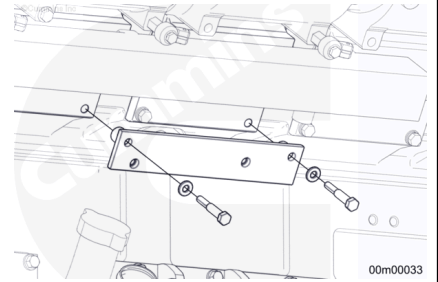
Tighten the 12 capscrews.

Torque Value: 40 n•m [30 ft-lb]



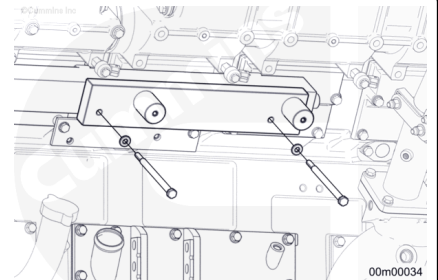
Install the crankcase breather relocation bracket. There are two capscrews, Cummins® Part Number S104, that will be installed into the forward holes of camshaft covers R4 and R5.

Torque Value: 40 n·m [30 ft-lb]



Install the debris shield bracket on top of the engine wire harness. There are two capscrews, Cummins® Part Number S170A, that will replace the existing wire harness capscrews for bracket installation.

Torque Value: 40 n·m [30 ft-lb]

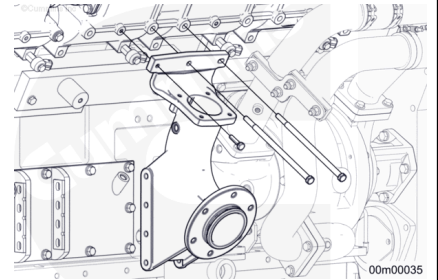


Install the fuel transfer connection onto the aftercooler housing between cylinders 3RB and 4RB. Use two capscrews, Cummins® Part Number 4968985, in the center and right holes.

Torque Value: 45 n·m [33 ft-lb]

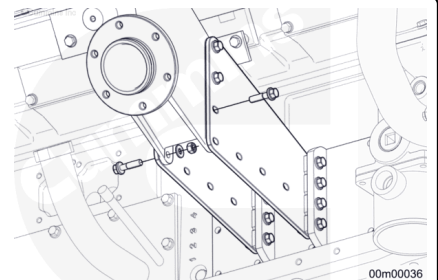
Use one capscrew, Cummins® Part Number S119, in the left hole.

Torque Value: 20 n·m [177 in-lb]



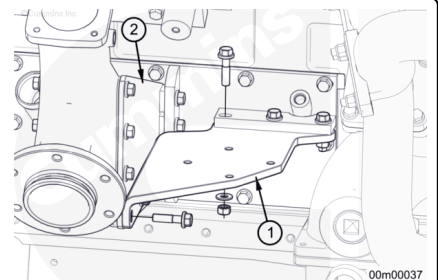
Install the fuel transfer connection support braces between the fuel transfer connection and hand hole cover. The rear support will have eight bolts (3033038), nuts (3201128), and washers (3042329). The forward support will have an identical seven bolts, nuts, and washers.

Torque Value: 40 n·m [30 ft-lb]



Install the new corrosion filter support bracket (1) between the forward support brace (2) and hand hole cover. Use eight bolts (3033038), nuts (3201128), and washers (3042329).

Torque Value: 40 n·m [30 ft-lb]



Install the coolant filter head using the original filter head capscrews. Leave hand-tight.

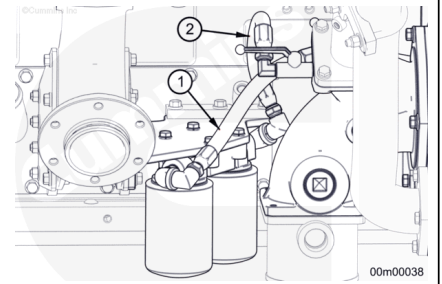
Install the coolant filter return (1) and supply lines (2).

Torque Value: 15 n•m [133 in-lb]

Tighten the four filter head capscrews.

Torque Value: 45 n•m [33 ft-lb]

Make sure that the filter supply and return ball valves are open upon completion of the coolant filter head installation.



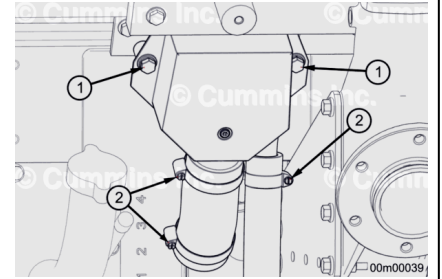
Install the crankcase breather onto the relocation bracket using the original crankcase breather capscrews. Align the hoses and hose clamps.

Tighten the crankcase breather capscrews (1).

Torque Value: 40 n•m [30 ft-lb]

Tighten the hose clamps (2).

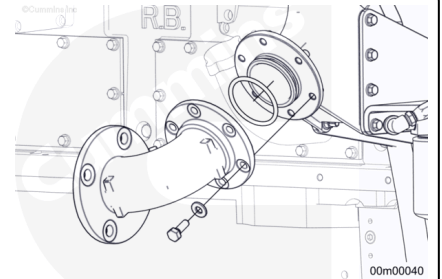
Torque Value: 6 n•m [53 in-lb]



The customer-to-engine fuel transfer connection can be oriented to the front or rear of the engine, depending on the OEM application. Consult with your local distributor or OEM to determine the correct direction for installation.

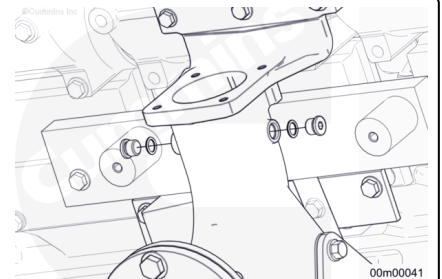
Align the customer-to-engine fuel transfer connection. Use six capscrews(S145), washers (3036900), and o-ring (4328686). Tighten the capscrews in a crisscross pattern.

Torque Value: 95 n•m [70 ft-lb]



Install both port plugs into the front and rear fuel transfer connection bungs.

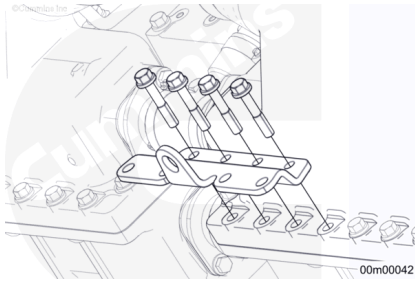
Torque Value: 45 n•m [33 ft-lb]



Fuel Control Valve

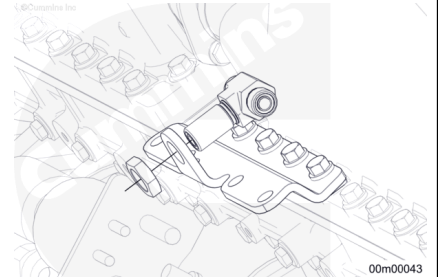
Install the debris shield bracket onto the corner of the aftercooler with four new capscrews, Cummins® Part Number 3040837.

Torque Value: 25 n•m [221 in-lb]



Install the male union tee onto the debris shield bracket.

Torque Value: 55 n·m [41 ft-lb]

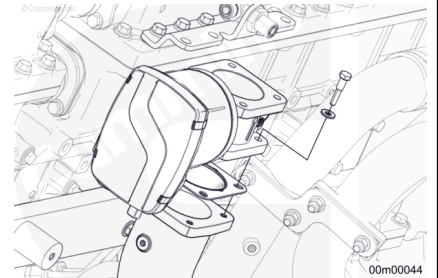


Place a gasket between the throttle body and the fuel transfer connection.

Install the throttle body with the arrow pointing upward, in the direction of gas flow.

Tighten the four capscrews, Cummins® Part Number S111.

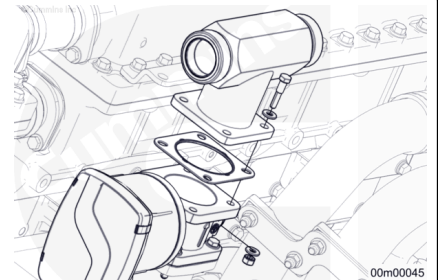
Torque Value: 22 n·m [195 in-lb]



Place a gasket between the throttle body and the fuel inlet connection tee.

Install four capscrews, Cummins® Part Number S111, connecting the fuel inlet connection tee to the throttle body.

Torque Value: 22 n·m [195 in-lb]

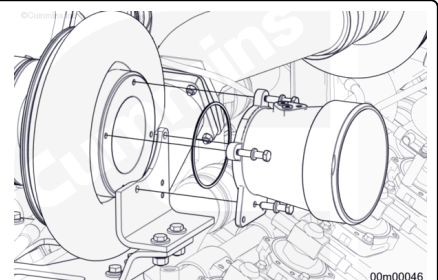


Air/Fuel Mixers

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb] or more. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Replace the existing turbocharger air intakes with the provided air/fuel mixers. The air/fuel mixers are installed using the same instructions for each bank.



Support the weight of the turbocharger so that it is secure upon removing the turbocharger air intake capscrews.

Remove the original low-pressure turbocharger air intakes. Three of the four air intake capscrews **must** be kept for reuse with the air/fuel mixers.

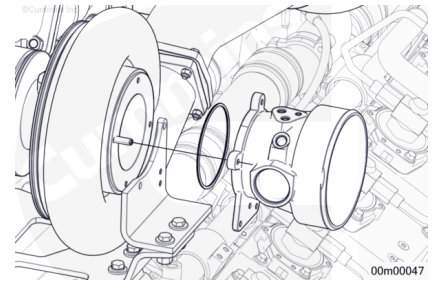
Remove and discard the o-ring on the back side of the turbocharger air intake.

Insert a new low-pressure turbocharger intake o-ring into the groove on the backside of the air/fuel mixer.

Before aligning the mixer, insert stud, Cummins® Part Number 4328541, into the bolt hole furthest to the left, when facing the turbocharger, and tighten.

Torque Value: 20 n·m [177 in-lb]

Align the stud hole with the air/fuel mixer. Rotate the air/fuel mixer into position. When facing the mixer, the gas inlet port **must** be oriented to the left



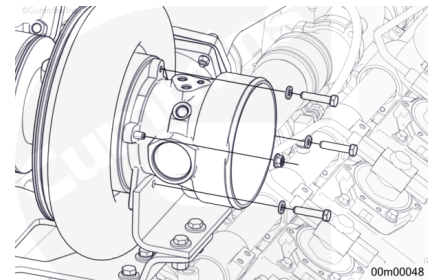
Install the three remaining capscrews through the air/fuel mixer into the compressor housing. Tighten the capscrews.

Torque Value: 48 n·m [35 ft-lb]

Install the nut onto the remaining stud and tighten.

Torque Value: 48 n·m [35 ft-lb]

Remove the turbocharger supports added during removal of the turbocharger air intake.



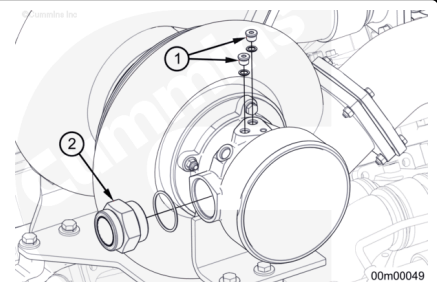
Make sure there is an o-ring between the male connector (2) and the air/fuel mixer. Install the male connector (2) into the air/fuel mixer using the crows foot adapter, Cummins® Part Number 5299067.

Torque Value: 331 n·m [244 ft-lb]

Note : The male connector (2) face with the enclosed o-ring should be oriented away from the air/fuel mixer.

Install the port plugs (1) into the dual fuel air/fuel mixer.

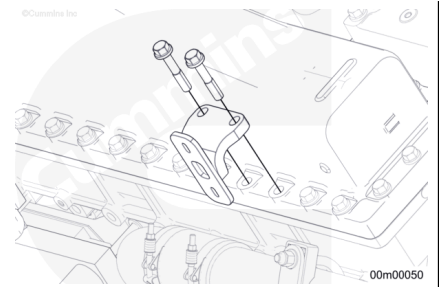
Torque Value: 45 n·m [33 ft-lb]



Left Bank Gas Inlet Tubes

Install the tube brace backing onto the front left aftercooler using capcrews, Cummins® Part Number 3040837. Discard the original 3rd and 4th capscrews back from the front corner of the aftercooler.

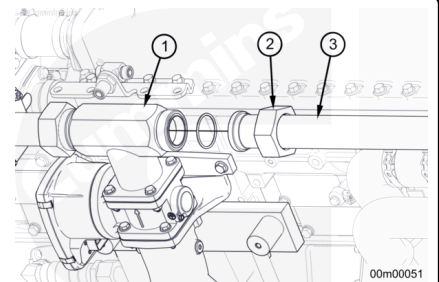
Torque Value: 25 n•m [221 in-lb]



Place o-ring, Cummins® Part Number 4327742, between the gas inlet tube (3) and the fuel inlet connection tee (1).

Install the gas inlet tube (3) on the fuel inlet connection tee (1) and tighten the connecting nut (2) using the crows foot adapter, Cummins® Part Number 5299067.

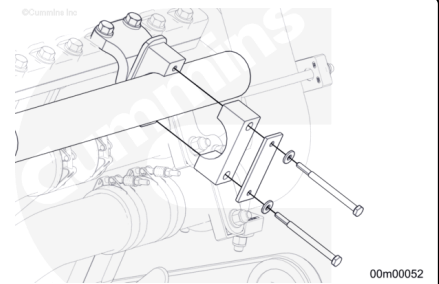
Torque Value: 508 n•m [375 ft-lb]



Make sure the horizontal gas inlet tube is positioned against both sections of the tube clamp on the aftercooler.

Install the tube clamp with capscrews, Cummins® Part Number 137719.

Torque Value: 11 n•m [97 in-lb]

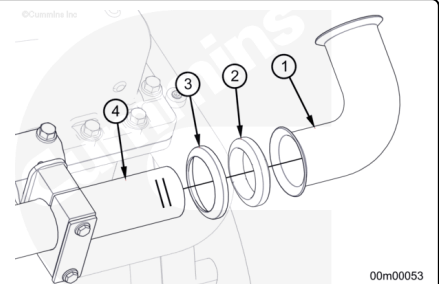


Draw or paint two reference marks on the horizontal gas inlet tube (4). One at 29 mm [1.14 in] and one at 42 mm [1.65 in] from the open end.

Install the metal clamp ring (3) over the tube (4). Slide a donut seal (2) onto the tube end (4) between the 29 mm [1.14 in] and 42 mm [1.65 in] marks.

Slide the tube connector elbow (1) over the horizontal gas inlet tube (4) until the 29 mm [1.14 in] mark is no longer visible.

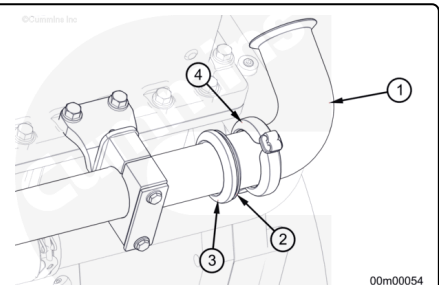
Note : Make sure the 42 mm [1.65 in] mark is still visible on the tube after assembly.



Make sure the donut seal (2) is positioned against the tube connector elbow (1) mounting surface.

Install the tube connector elbow clamp (4) over the tube connector elbow (1), donut seal (2), and metal clamp ring (3). Tighten.

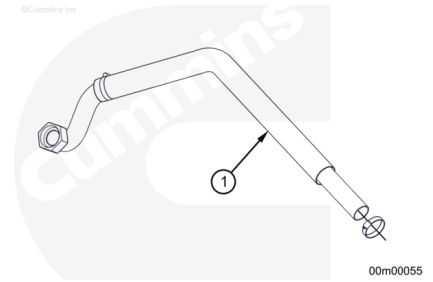
Torque Value: 11 n•m [97 in-lb]



Install the heat shield (1) onto the upper gas inlet tube.

Tighten the hose clamps, Cummins® Part Number, 43828D.

Torque Value: 6 n·m [50 in-lb]

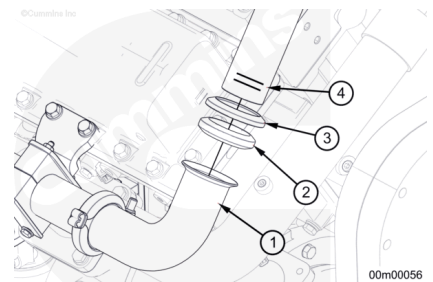


Draw or paint two reference marks on the upper gas inlet tube (4) - One at 29 mm [1.14 in] and one at 42 mm [1.65 in] from the open end.

Install the metal clamp ring (3) over the tube (4). Slide a donut seal (2) onto the tube between the 29 mm [1.14 in] and 42 mm [1.65 in] marks.

Slide the upper gas inlet tube (4) into the tube connector elbow (1) until the 29 mm[1.14 in] mark is no longer visible.

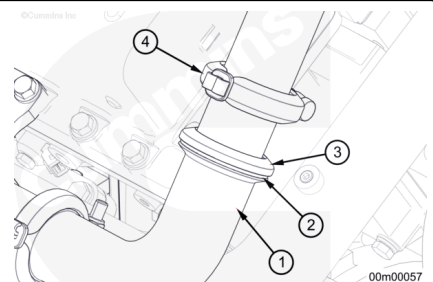
Note : Make sure the 42 mm [1.65 in] mark is still visible on the tube after assembly.



Make sure the donut seal (2) is positioned against the tube connector elbow mounting surface.

Install the tube connector elbow clamp (4) over the tube connector elbow (1), donut seal (2), and metal clamp ring (3). Tighten.

Torque Value: 11 n·m [97 in-lb]

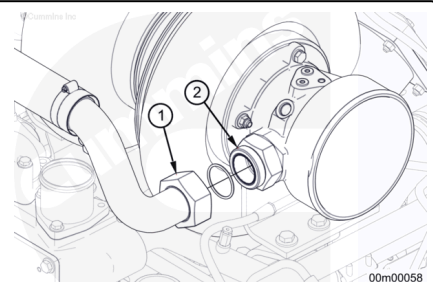


Install an o-ring between the gas inlet tube connecting nut (1) and the male connector (2) at the air/fuel mixer.

Tighten the connecting nut using the crows foot adapter, Cummins® Part Number 5299067.

Note : A second wrench must be used to stabilize the male connector on the mixer.

Torque Value: 331 n·m [244 ft-lb]

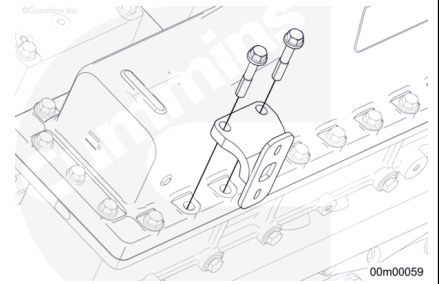


Right Bank Gas Inlet Tubes

Install two tube brace backings onto the rear left aftercooler using capcrews, Cummins® Part Number 3040837, onto the first and second capscrew holes from the aftercooler front

corner and the second and third capscrew holes from the aftercooler rear corner.

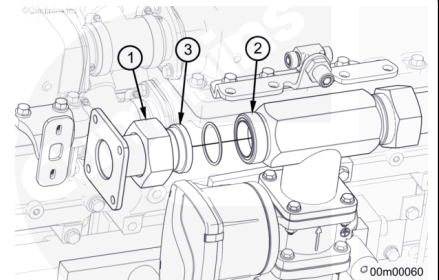
Torque Value: 25 n•m [221 in-lb]



Place o-ring, Cummins® Part Number 4327742, between the gas inlet tube (3) and the fuel inlet connection tee (2).

Install the gas inlet tube (3) on the fuel inlet connection tee (2) and tighten the connecting nut (1) using the crows foot adapter, Cummins® Part Number 5299067.

Torque Value: 508 n•m [375 ft-lb]

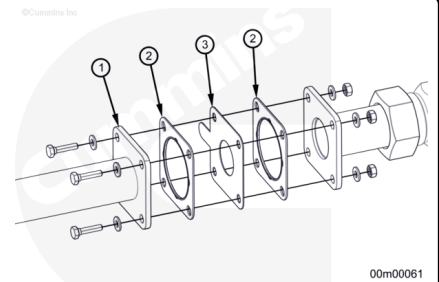


Install gaskets (2) on either side of the orifice plate (3).

Install the horizontal gas inlet tube (1).

Install and tighten the four bolts (S119), washers (3042328), and nuts (69832) on the orifice plate assembly.

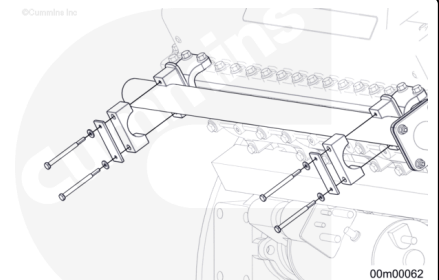
Torque Value: 38 n•m [28 ft-lb]



Make sure the horizontal gas inlet tube is positioned against both tube clamps on the aftercooler.

Install the tube clamp with capscrews, Cummins® Part Number 137719.

Torque Value: 11 n•m [97 in-lb]

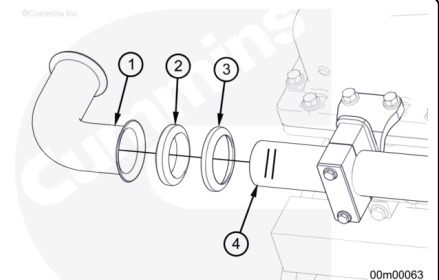


Draw or paint two reference marks on the horizontal gas inlet tube (4) - One at 29 mm [1.14 in] and one at 42 mm [1.65 in] from the open end.

Install the metal clamp ring (3) over the tube (4). Slide a donut seal (2) onto the tube end (4) between the 29 mm [1.14 in] and 42 mm [1.65 in] marks.

Slide the tube connector elbow (1) over the horizontal gas inlet tube (4) until the 29 mm [1.14 in] mark is no longer visible.

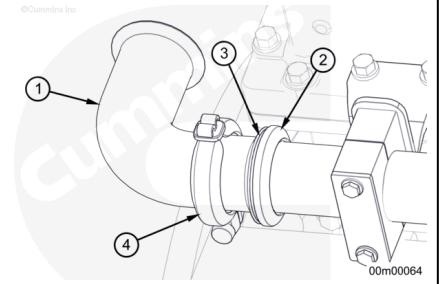
Note : Make sure the 42 mm [1.65 in] mark is still visible on the tube after assembly.



Make sure the donut seal (2) is positioned against the tube connector elbow (1) mounting surface.

Install the tube connector elbow clamp (4) over the tube connector elbow (1), donut seal (2), and metal clamp ring (3). Tighten.

Torque Value: 11 n·m [97 in-lb]

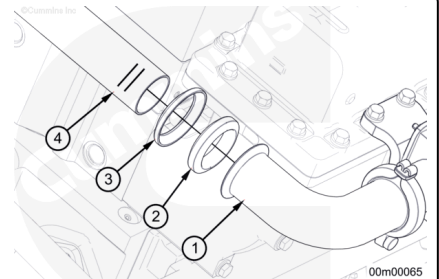


Draw or paint two reference marks on the upper gas inlet tube (4). One at 29 mm [1.14 in] and one at 42 mm [1.65 in] from the open end.

Install the metal clamp ring (3) over the tube (4). Next slide a donut seal (2) onto the tube between the 29 mm [1.14 in] and 42 mm marks.

Slide the upper gas inlet tube (4) into the tube connector elbow (1) until the 29 mm [1.14 in] mark is no longer visible.

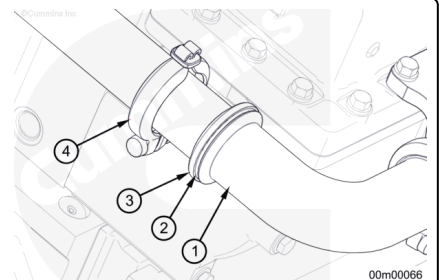
Note : Make sure the 42 mm [1.65 in] mark is still visible on the tube after assembly.



Make sure the donut seal (2) is positioned against the tube connector elbow mounting surface.

Install the tube connector elbow clamp (4) over the tube connector elbow (1), donut seal (2), and metal clamp ring (3). Tighten.

Torque Value: 11 n·m [97 in-lb]

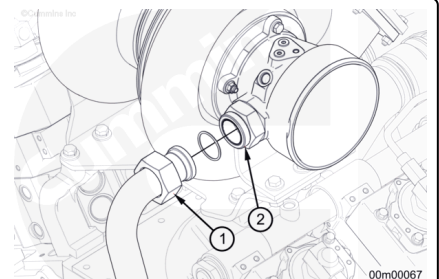


Make sure an o-ring is between the gas inlet tube connecting nut (1) and the male connector (2) at the air/fuel mixer.

Tighten the connecting nut using the crows foot adapter, Cummins® Part Number 5299067.

Note : A second wrench must be used to stabilize the male connector on the mixer.

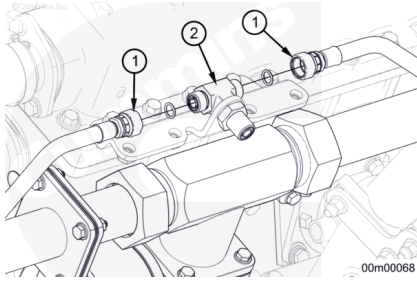
Torque Value: 331 n·m [244 ft-lb]



Fuel Regulator Air Pressure Signal Hose

Install the left and right bank fuel regulator air pressure signal hose (1) onto the T-fitting (2) at the throttle valve debris shield.

Torque Value: 54 n·m [40 ft-lb]



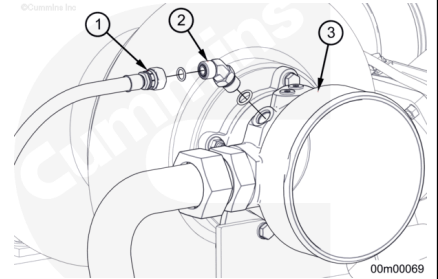
Install the 45 degree male fittings (2) into the left and right bank air/fuel mixers (3).

Torque Value: 54 n·m [40 ft-lb]

Install the fuel regulator air pressure signal hose (1) onto the left and right bank 45 degree male fittings (2). Hand-tighten.

Make sure the fitting is oriented towards the main gas manifold tube.

Torque Value: 54 n·m [40 ft-lb]



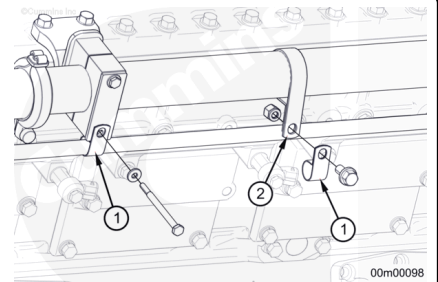
The left and right bank fuel regulator air pressure signal hoses (1) will be suspended from P-clips (1) that are supported from the gas manifold tubes.

Each gas manifold tube clamp needs one P-clip (1), Cummins® Part Number 69911A, added to the bottom capscrew for fuel regulator air pressure signal hose support.

Torque Value: 11 n·m [97 ft-lb]

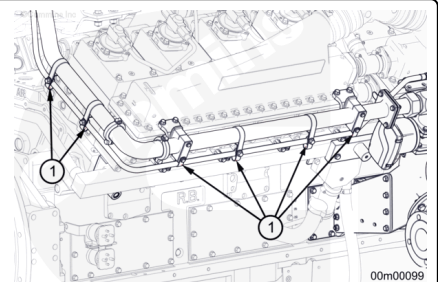
Several larger P-clips (2), Cummins® Part Number 3202124, will be attached to the gas tubes with nut, Part Number 3064321, and bolt, Part Number 3175239, for fuel regulator air pressure signal hose support.

Torque Value: 41 n·m [30 ft-lb]



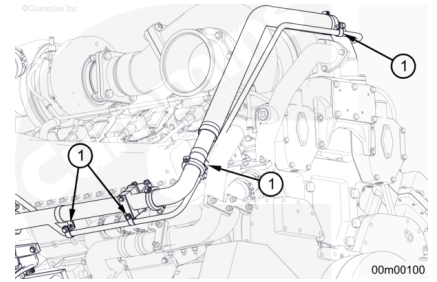
P-clip location for right bank rear fuel regulator air pressure signal hose support (1).

Note : Failure to completely secure the fuel regulator air pressure signal hose may result in premature hose wear.



P-clip location for right bank front fuel regulator air pressure signal hose support (1).

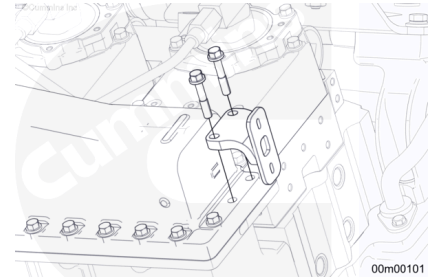
Note : Note: Failure to completely secure the fuel regulator air pressure signal hose may result in premature hose wear.



On-Engine Wiring Harness Bracket

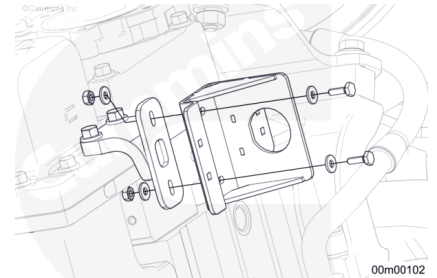
Install the wiring harness bracket mounting place onto the rear left-bank aftercooler using capscrews, Cummins® Part Number 3040837.

Torque Value: 25 n•m [221 in-lb]



Install the wire harness bracket onto the mounting plate using capscrews, Cummins® Part Number 4087931.

Torque Value: 11 n•m [97 ft-lb]

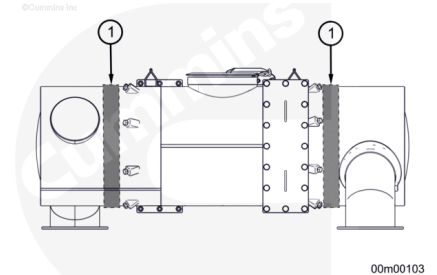


Diesel Oxidation Catalyst

Proper mounting and installation of the diesel oxidation catalyst (DOC) canister must be planned and organized through the OEM and Cummins® Distribution prior to final assembly. The following instructions will guide the technician through canister installation after initial canister mounting planning and fabrication has been completed.

The DOC aftertreatment **must only** have brackets/mounting straps located in designated mounting zones (1). Prior to starting the installation process, use approved mounting zones to determine the mount attachment point on the support frame.

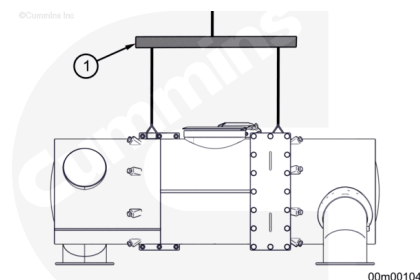
Note : For a complete list of aftertreatment mounting and installation requirements, please consult with the Cummins® Distributor and/or OEM.



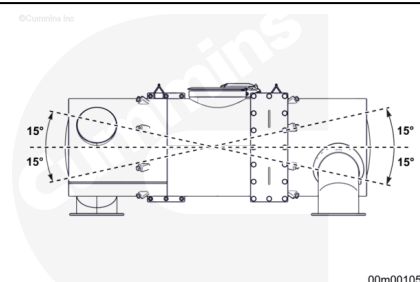
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

The DOC lifting brackets are designed to lift the DOC canister **only**. Lifting other components that are attached to the DOC can cause damage to the DOC assembly.

Note : When lifting the canister, adjust the strap angle to be less than 45 degrees from vertical. A spreader bar (1) is preferred.



DOC installation angularity should be $0^\circ \pm 15^\circ$ in all directions.



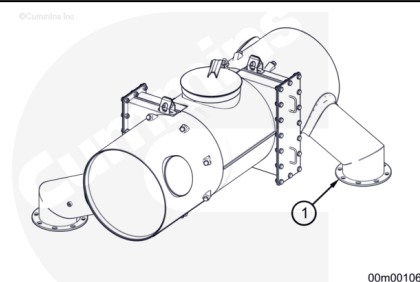
Using the supplied gaskets, attach the DOC inlet flanges (1) to the OEM-supplied exhaust piping.

Fasteners are to be supplied by either the OEM and/or Cummins® Distributor. Tighten the fasteners in a crisscross pattern.

Torque Value: 129 n·m [95 ft-lb]

Tighten the fasteners in a crisscross pattern.

Torque Value: 265 n·m [196 ft-lb]

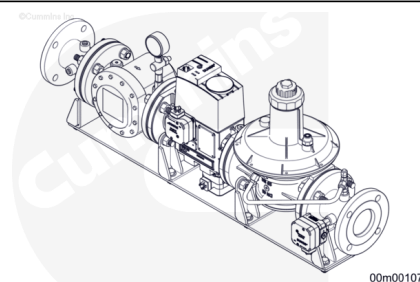


Off-Engine Gas Train

General Information

The following instructions detail how to assemble the Dual Fuel Off-Engine Gas Train using all Cummins® factory-supplied components in the Dual Fuel Kit.

Actual assembly and mounting of the gas train may differ, depending on OEM and/or Distributor guidelines. Consult the OEM/Distributor for proper installation and mounting instructions specific to the application.



The gas train **must** be assembled in the following order: filter, fuel shutoff valve, and regulator.

Note : Gas train components **must** be inspected for seals, plastic covers, and/or end caps over component orifices during installation. Remove all seals, plastic covers, and/or end caps from the components before assembly.

Note : The fuel shutoff valve comes with a closure validation switch already installed onto the bottom of the valve 2 housing. Do not set the fuel shutoff valve on top of this switch.

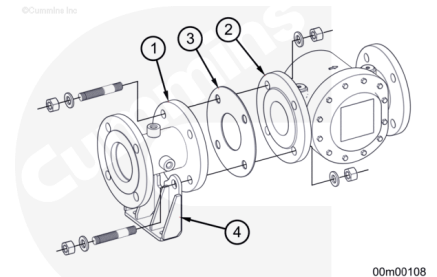
Align the OEM flange (1) and the fuel filter housing (2). Insert a gasket, Cummins® Part Number 4322911 (3), between the components. Make sure the fuel filter housing arrow is pointing downstream.

Insert studs, Part Number 4327792, through the four flange holes. Align the mounting bracket (4) through the studs on the outside of the OEM flange (1). Add washers and nuts to each side of the studs.

Note : NOTE: The OEM flange has two ports. It must be oriented so one port faces up and the other port faces outward, away from the engine.

Note : There are two different bolt patterns on the OEM flange component. Make sure the holes align between the OEM flange and the filter housing.

Note : The fuel filter should be oriented so the filter cover is pointing outward, away from the engine. Consult the OEM/Distributor if a different orientation is required.

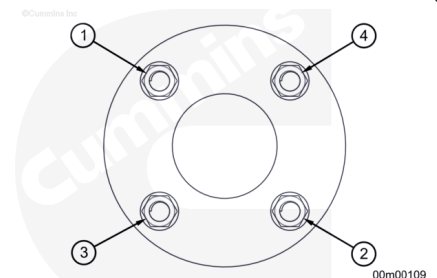


Note : The gas train components are comprised of raised face flanges. Care must be taken to make sure that the flanges are aligned flush against each other during assembly.

Bring the flanges together. Make sure that a new gasket has been placed between the flanges.

With the studs, washers, and nuts in place, hand-tighten each of the four bolted joints so that the raised faces are seated evenly against one another.

Tightening of the nuts **must** be done in three steps.



Tighten the nuts in a crisscross pattern.

Torque Value: 16 n•m [142 in-lb]

Tighten the nuts in a crisscross pattern.

Torque Value: 33 n•m [24 ft-lb]

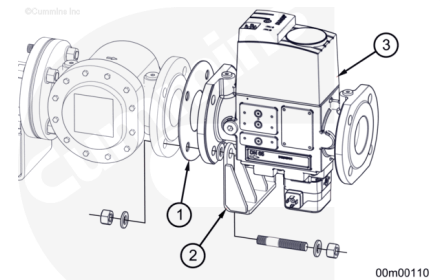
Tighten the nuts in a crisscross pattern.

Torque Value: 50 n•m [37 ft-lb]

Align the fuel shutoff valve (3) and the fuel filter housing. Insert a gasket, Cummins® Part Number 4322911 (1) between the components. Make sure the fuel shutoff valve (3) housing arrow is pointing downstream.

Install the studs (4327792) through the fuel filter flanges into the fuel shutoff valve (3). Install the mounting bracket and washers onto the studs on the fuel shutoff valve side.

Note : There should be washers on both sides of the fuel shutoff valve mounting foot.



Note : The gas train components are comprised of raised face flanges. Care must be taken to make sure that the flanges are aligned flush against one another during assembly.

Bring the flanges together. Make sure that a new gasket has been placed between the flanges.

With the studs, washers, and nuts in place, hand-tighten each of the four bolted joints so that the raised faces are seated evenly against one another.

Tightening of the nuts must be done in three steps.

Tighten the nuts in a crisscross pattern.

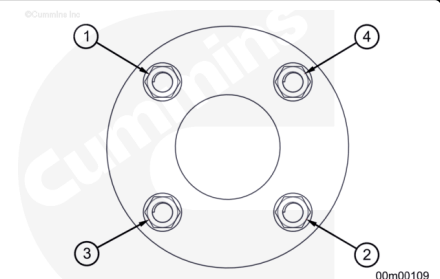
Torque Value: 16 n•m [142 in-lb]

Tighten the nuts in a crisscross pattern.

Torque Value: 33 n•m [24 ft-lb]

Tighten the nuts in a crisscross pattern.

Torque Value: 50 n•m [37 ft-lb]

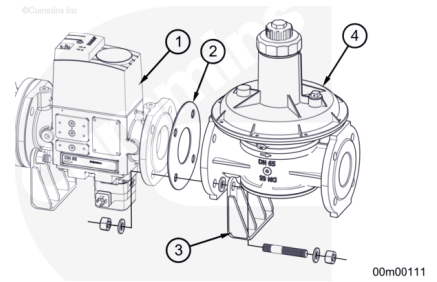


Align the pressure regulator (4) and the fuel shutoff valve (1). Insert a gasket, Cummins® Part Number 4322911 (2), between the components. Make sure the pressure regulator housing arrow is pointing downstream.

Insert studs, Cummins® Part Number 4327792, through the four flange holes

Install the mounting bracket (3) and washers onto the studs on the pressure regulator side.

Note : There should be washers on both sides of the pressure regulator mounting foot.



Note : The gas train components are comprised of raised face flanges. Care must be taken to make sure that the flanges are aligned flush against one another during assembly.

Bring the flanges together. Make sure that a new gasket has been placed between the flanges.

With the studs, washers, and nuts in place, hand-tighten each of the four bolted joints so that the raised faces are seated evenly against each other.

Tightening of the nuts **must** be done in three steps.

Tighten the nuts in a crisscross pattern.

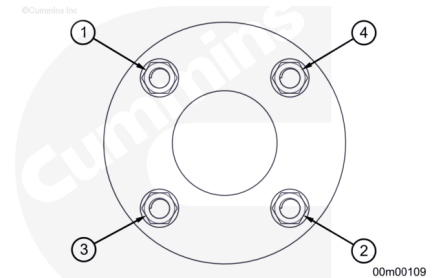
Torque Value: 16 n·m [142 in-lb]

Tighten the nuts in a crisscross pattern.

Torque Value: 33 n·m [24 ft-lb]

Tighten the nuts in a crisscross pattern.

Torque Value: 50 n·m [37 ft-lb]

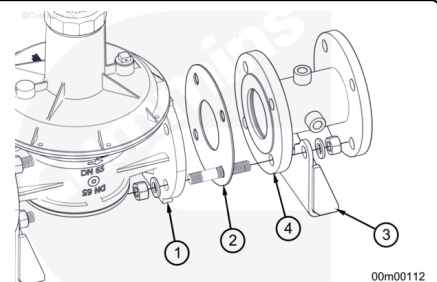


Align the pressure regulator (1) and the OEM flange (4). Insert a gasket, Cummins® Part Number 4322911 (2), between the components.

Insert studs, Cummins® Part Number 4327792, through the four flange holes. Align the mounting bracket (3) through the studs on the outside of the OEM flange (4). Add washers and nuts to each side of the studs.

Note : The OEM flange has two ports, it must be oriented so one port faces up and the other port faces outward, away from the engine.

Note : There are two different bolt patterns on the OEM flange component. Make sure the holes align between the OEM flange and the fuel regulator.



Note : The gas train components are comprised of raised face flanges. Care must be taken to make sure that the flanges are aligned flush against one another during assembly.

Bring the flanges together. Make sure that a new gasket has been placed between the flanges.

With the studs, washers, and nuts in place, hand-tighten each of the four bolted joints so that the raised faces are seated evenly against each other.

Tightening of the nuts **must** be done in three steps.

Tighten the nuts in a crisscross pattern.

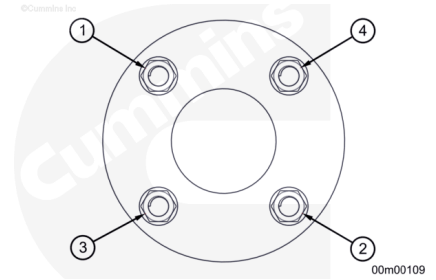
Torque Value: 16 n•m [142 in-lb]

Tighten the nuts in a crisscross pattern.

Torque Value: 33 n•m [24 ft-lb]

Tighten the nuts in a crisscross pattern.

Torque Value: 50 n•m [37 ft-lb]

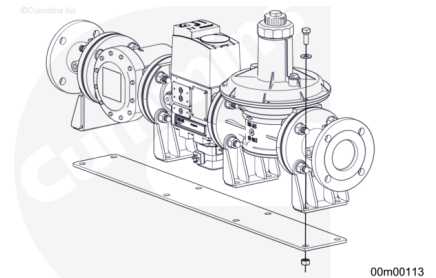


Align the gas train assembly mounting brackets on top of the gas train mounting plate.

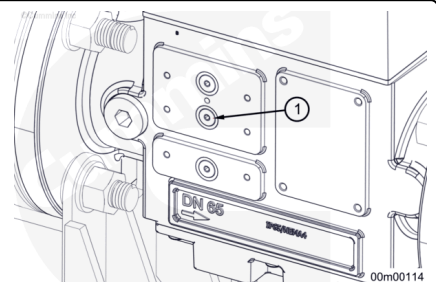
Use mounting bolts, Cummins® Part Number 145, and washers to assemble the feet to the plate.

Torque Value: 95 n•m [70 ft-lb]

Note : The final placement and location of the gas train assembly onto the trailer application will be determined by the OEM/Distributor.



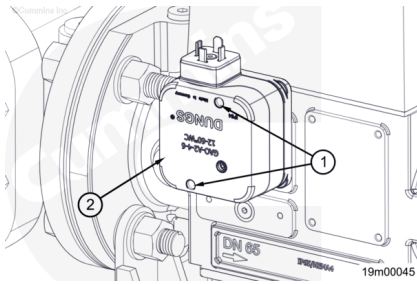
New fuel shutoff valves come with port plugs. Before installing the low-pressure switch, make sure the port plug in port 2 (1) on the fuel shutoff valve is removed.



Remove and keep the two screws (1) from the clear cover on the low-pressure switch.

Remove the clear cover (2) from the low-pressure switch.

Inside the pressure switch will be two capscrews and an o-ring to mount the switch to port 2 on the fuel shutoff valve.



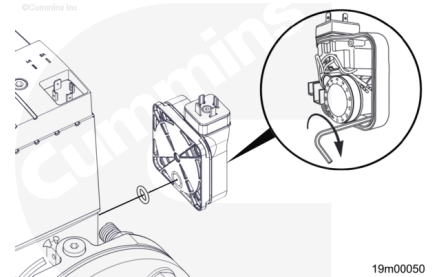
Place an o-ring in the port groove on the back of the low-pressure switch.

Align the back of the switch with the number 2 port hole on the fuel shutoff valve. Make sure the o-ring is seated against the fuel shutoff valve and the screw holes are aligned.

Verify the low-pressure switch is being installed on fuel shutoff valve port number 2 and the switch connector is oriented upwards.

Insert the two capscrews through the low-pressure switch body into the fuel shutoff valve and tighten.

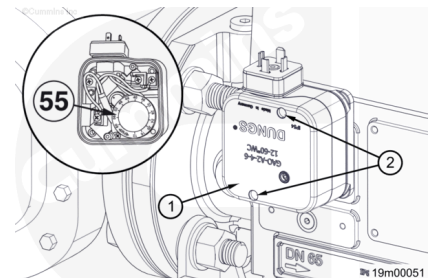
Torque Value: 2.5 n·m [22 in-lb]



Make sure the low pressure switch dial is set to 55 in H₂O.
Install the clear cover on the low pressure switch.

Insert screws and tighten.

Torque Value: 0.7 n·m [6 in-lb]



Install the pulse line tee fitting into the downstream OEM flange.

Male connector (3)

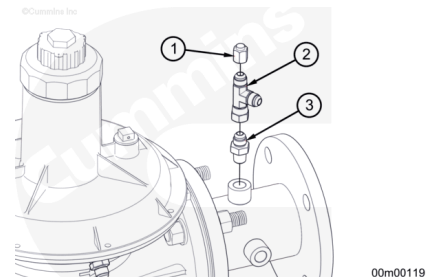
Torque Value: 47 n·m [35 ft-lb]

Install the tee fitting (2) hand-tight. Align the tee fitting so that it is angled for installation of the pulse line. Tighten.

Torque Value: 47 n·m [35 ft-lb]

Cap (1)

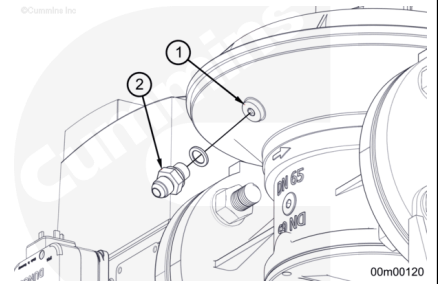
Torque Value: 11 n·m [97 in-lb]



Remove the existing port plug (1) from underneath the pressure regulator diaphragm.

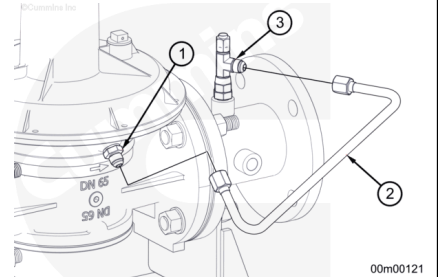
Install the regulator male connector (2).

Torque Value: 23 n•m [204 in-lb]



Install the pulse line (2) between the tee fitting (3) and regulator male connector (1).

Torque Value: 18 n•m [159 ft-lb]

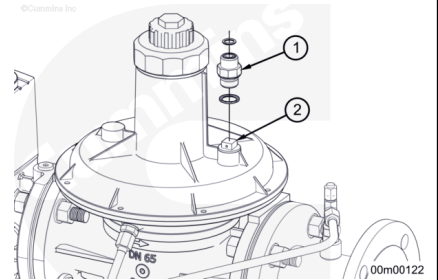


Remove the existing port plug (2) from the top of the pressure regulator diaphragm.

Install the male-male fuel regulator air pressure hose line fitting (1).

Torque Value: 50 n•m [37 ft-lb]

Note : The fuel regulator air pressure signal line from the off-engine gas train to the on-engine components is provided by either the distributor or OEM. Consult your OEM/Distributor for the proper installation guidelines of the fuel regulator air pressure hose line.

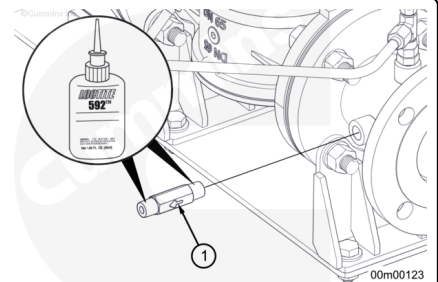


Apply Loctite™ 592, or equivalent gas compatible thread sealant, to the bare threads on both sides of the hex orifice fitting.

Install the fitting into the downstream OEM flange pressure port facing away from the engine.

Torque Value: 20 n•m [177 in-lb]

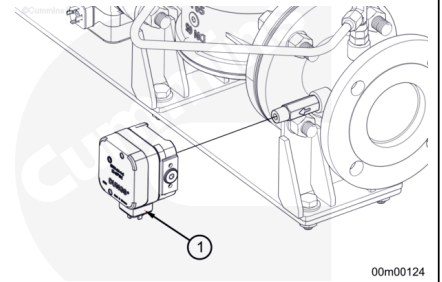
Note : The hex orifice fitting must be installed so that the arrow (1) points toward the pressure switch, away from the pipe.



Thread the back of the high-pressure switch onto the hex fitting.

Torque Value: 20 n•m [180 in-lb]

Note : Make sure the connector (1) is oriented downward upon installation.



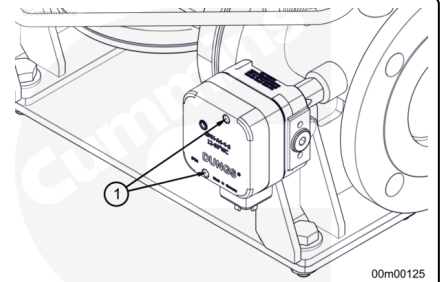
Make sure the high-pressure switch is set to 35 in H₂O.

If the switch is **not** set to 35 inH₂O, remove the two screws (1) from the high-pressure switch front cover.

Set the switch to 35 inH₂O by turning the dial.

Install the high-pressure switch cover.

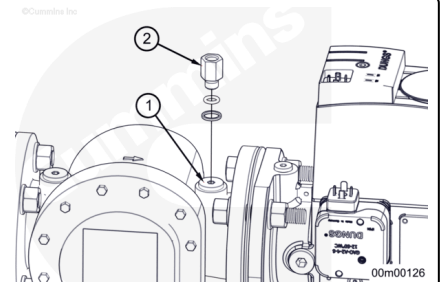
Torque Value: 0.7 n•m [6 in-lb]



Remove the OEM threaded plug (1).

Install the threaded adapter, Cummins® Part Number 4322908 (2), o-ring, and o-ring retainer into the downstream fuel filter housing port.

Torque Value: 18 n•m [159 in-lb]



Apply Loctite™ 592, or equivalent gas compatible thread sealant, to both ends of the hex fitting Cummins® Part Number 4327787 (1).

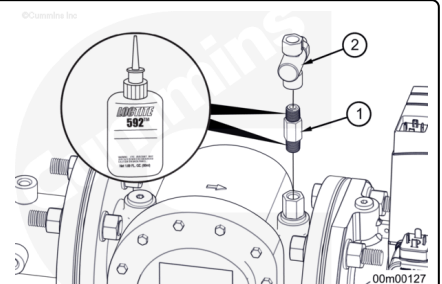
Install the hex fitting onto the threaded adapter and tighten.

Note : Use a second wrench to stabilize the threaded adapter while tightening, to avoid over tightening

Torque Value: 20 n•m [177 in-lb]

Install the push button valve (2) onto the hex fitting and tighten.

Note : Use a second wrench to stabilize the hex fitting while tightening, to avoid overtightening of the threaded adapter.

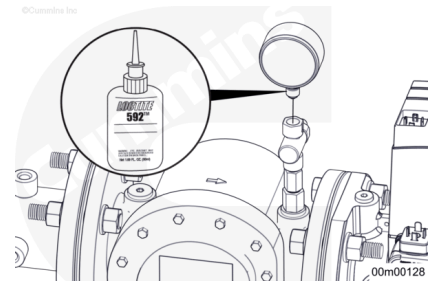


Apply Loctite™ 592, or equivalent gas compatible thread sealant, to the threaded end of the pressure gauge.

Install the pressure gauge into the push button valve.

Torque Value: 20 n•m [177 in-lb]

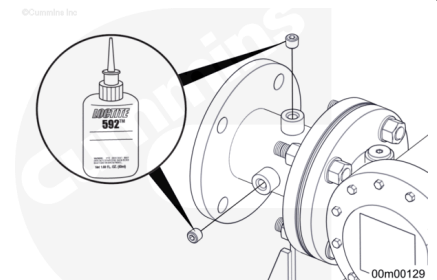
Note : Make sure the pressure gauge faces the operator. Use a second wrench to stabilize the push button valve while tightening to avoid overtightening of the threaded adapter.



Apply Loctite™ 592, or equivalent gas compatible thread sealant, to the two port plug threads.

Install both port plugs into the OEM flange test ports.

Torque Value: 20 n•m [177 in-lb]

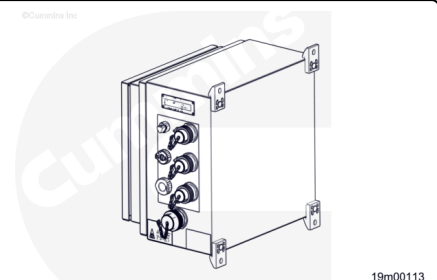


Control Panel

General Information

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb] or more. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



There are four dual fuel panel mounting brackets; one at each corner.

Mounting hardware and installation location guidelines are both provided by either the OEM or Cummins® Distributor. Consult the OEM and/or installing Cummins® Distributor for control panel mounting instructions.

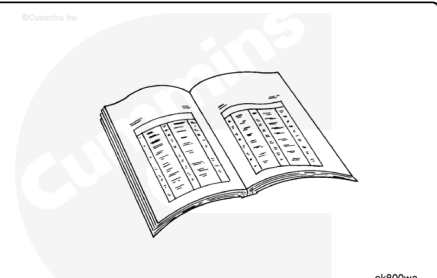
Dual Fuel System Wire Harnesses

General Information

The QSK50G Dual Fuel wiring harness is comprised of seven main harnesses. The following procedure details installation instructions for the Cummins® factory-supplied harnesses.

Cummins Factory supplied:

1. Battery and Ground
2. On-Engine
3. On-Engine to Panel



4. Gas Train
5. DOC

OEM supplied:

1. Hydraulic pump harness (speed and pressure input)
2. Fracturing trailer harness (4-20mA hydraulic horsepower signal and OEM gas shutoff input)

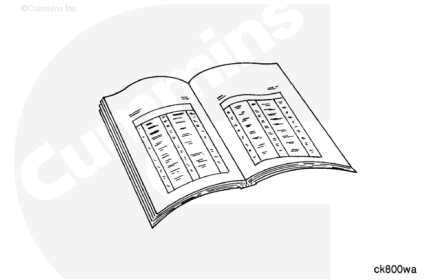
Note : Not all OEMs require the use of the OEM-supplied harnesses.

The OEM and/or Distributor will provide installation instructions for non-Cummins® factory-supplied wire harnesses. Consult the OEM/Distributor for proper hydraulic pump harness and fracturing trailer harness installation guidelines if applicable.

All harnesses, when installed, **must** satisfy the following requirements:

- No interference with machine components or any existing connectors
- **Must** have adequate clearance for removal of connectors from mating parts (sensors, actuators, solenoids, etc.)
- Clearance from rotating and hot components
- No potential points where the harness could rub
- No harness stretching to reach its respective mating ends
- **Must** be mechanically supported and secured
- At the Dual Fuel control panel, harnesses will be physically supported by use of tie wrap to minimize the bend radius of the connector wires, limiting physical degradation over the lifetime of this product.

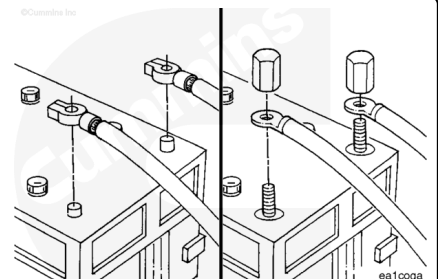
Note : All connectors are physically labeled on the wire harnesses.



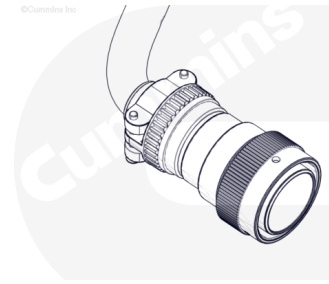
Battery and Ground

Consult with the OEM/Cummins Distributor for guidelines on how to connect the power supply harness for the Dual Fuel control module.

Connect the four 24 VDC power supply cables (red) to the positive lead (+) of the battery and connect the four common ground cables (black) to the negative lead (-) of the battery.



Connect the power supply wiring harness connector C2 to the Dual Fuel panel port C1.

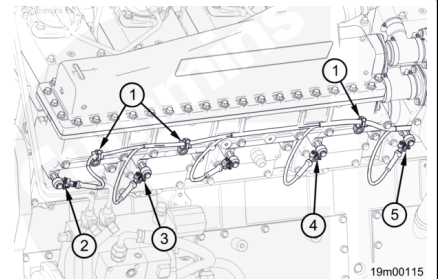


19m00114

On-Engine

Left Bank

Beginning with the left front quadrant, install the on-engine harness to the engine using three tie mounts (1) with capscrew, Cummins® Part Number S107.

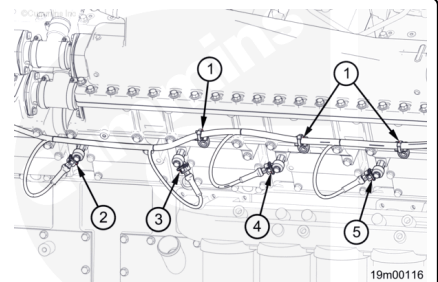


19m00115

Torque Value: 20 n·m [177 in-lb]

Connect the 1LB (2), 2LB (3), 3LB (4), and 4LB (5) knock sensors to connectors C22-1 (2), C22-3 (3), C22-5 (4), C22-7 (5).

On the left rear quadrant, install the on-engine harness to the engine using three tie mounts (1) with capscrew, Cummins® Part Number S107.

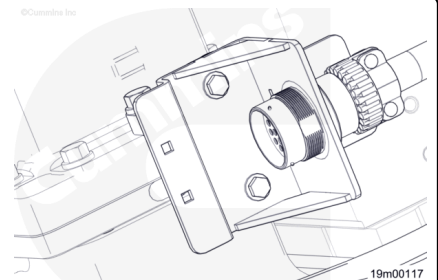


19m00116

Torque Value: 20 n·m [177 in-lb]

Connect the 5LB (2), 6LB (3), 7LB (4), and 8LB (5) knock sensors to connectors C22-9 (2), C22-11 (3), C22-13 (4), C22-15 (5)

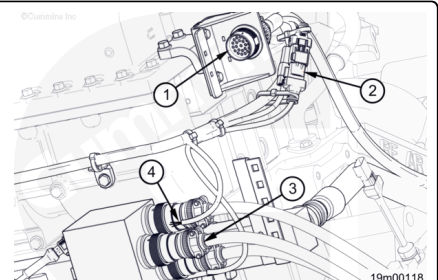
Support the main on-engine wire harness connector C10 at the corner of the rear left bank aftercooler. Tighten the connector locknut.



19m00117

Torque Value: 40 n·m [30 ft-lb]

Near the main on-engine wire harness connector (1), make sure connector C27 is mated to connector C28 (2) of the table switching output stub (4).

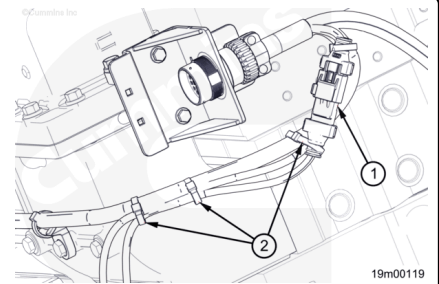


19m00118

The table switching output stub (4) includes three pins which **must** be added to the existing OEM 23 pin connector (3) that is a part of the QSK50 base engine. Each pin wire is labeled with the correct position to be added to the connector.

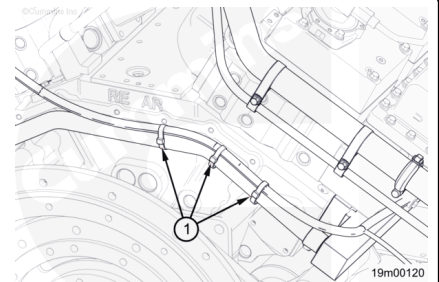
Also included in the on-engine harness install is the tachometer output signal stub (1). Make sure harness connector C21-2 is mated to stub connector C21-3 for the tachometer output signal.

The tachometer output signal stub contains two bare pins, which are sealed in heat shrink wrap. Leave the shrink wrap over the pins. The tachometer output signal stub is currently **not** used in the Dual Fuel System operation. Secure the tachometer output sensor stub with tie wrap (2) so that it does **not** interfere with Dual Fuel operation.



Rear

Secure the on-engine Dual Fuel harness to the existing QSK50 base engine harness across the top of the flywheel housing using tie wraps (1).



Right Bank

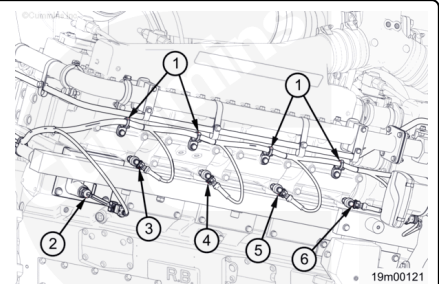
On the right rear quadrant, install the on-engine harness to the engine using four tie mounts (1) with capscrew, Cummins® Part Number S107.

Torque Value: 20 n·m [177 in-lb]

Connect the Dual Fuel camshaft position sensor to connector C21-1 (2).

Note : The camshaft sensor harness connector comes with a cap. Store this cap in a secure location.

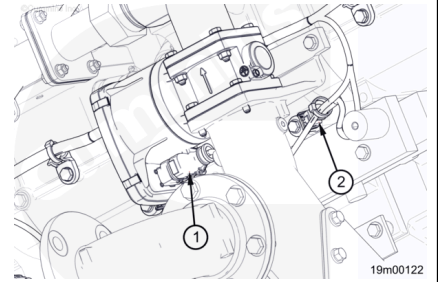
Connect the 8RB (3), 7RB (4), 6RB (5), and 5RB (6) knock sensors to connectors C22-16 (3), C22-14 (4), C22-12 (5), and C22-10 (6).



The wire harness **must** be routed behind the throttle valve assembly.

Connector C23 (1) plugs into the throttle valve.

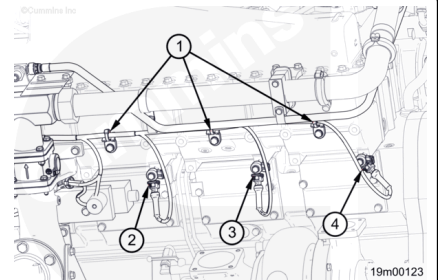
Connect the 4RB knock sensor (2) to connector C22-8 (2) located behind the fuel transfer connection.



On the right front quadrant, install the on-engine harness to the engine using three tie mounts (1) with capscrew, Cummins® Part Number S107.

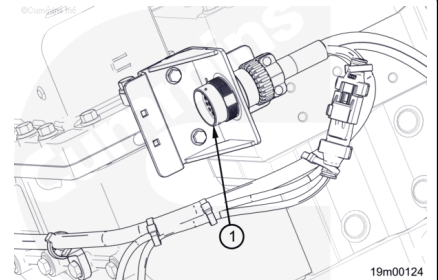
Torque Value: 20 n·m [177 in-lb]

Connect the 3RB (2), 2RB (3), and 1RB (4) knock sensors to connectors C22-6 (2), C22-4 (3), and C22-2 (4)

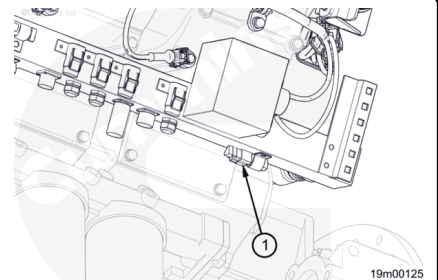


On-Engine to Panel Harness

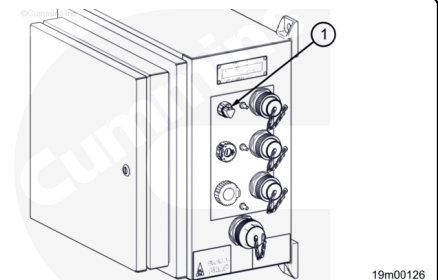
Connect the on-engine harness connector C10 (1) to the on-engine to Dual Fuel panel harness connector C9.



Mate the data link connector C20-2 (1) to the existing QSK50 base engine mounted OEM harness connector DT06-3S.



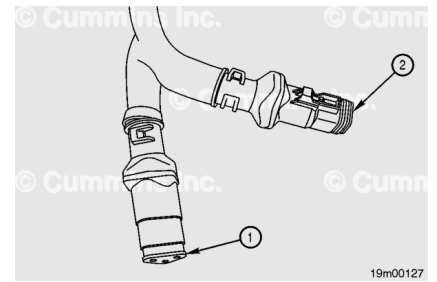
Connect the data link panel connector C6 to the Dual Fuel panel port C5 (1).



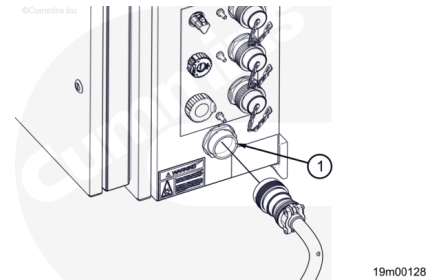
The data link OEM connector (2 - with blue tip) and the data link panel connector (1 - with orange tip) are identical in appearance. Close attention should be given to make sure each

connector is correctly installed. They can be distinguished by the identifying tags attached just above the connector.

Mate the data link OEM connector C20-1 (2) to the OEM J1939 connection. The OEM J1939 connection **must** be relocated to the ComAp panel.



Connect the on engine to panel harness connector C8 to the Dual Fuel panel port C7 (1).



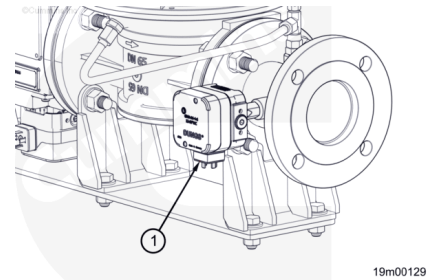
Gas Train

Make sure there is a rubber gasket and that there are no bent pins on the male high-pressure fuel switch connector. Install a new gasket, if necessary.

Align and connect C18-2 to the high-pressure switch (1) on the downstream OEM flange.

Install the captive screw and tighten.

Torque Value: 0.7 n•m [6 in-lb]



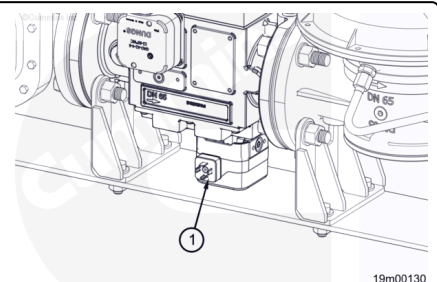
Make sure there is a rubber gasket and that there are no bent pins on the male closure validation switch connector. Install a new gasket, if necessary.

Align and connect C29-2 to the POC switch (1).

Install the captive screw and tighten.

Torque Value: 0.7 n•m [6 in-lb]

Note : There is an additional wire wrapped in heat shrink branching off from the POC switch wiring harness connector. Do not alter this wire.

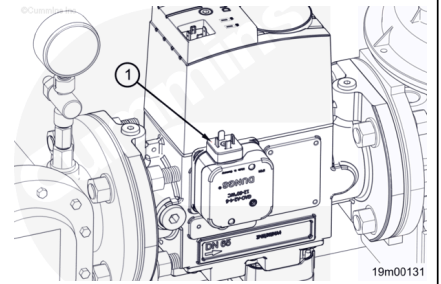


Make sure there is a rubber gasket and that there are no bent pins on the male low pressure fuel switch connector. Install a new gasket if necessary.

Align and connect C18-1 to the low pressure switch (1) on the fuel shutoff valve.

Install the captive screw and tighten.

Torque Value: 0.7 n•m [6 in-lb]

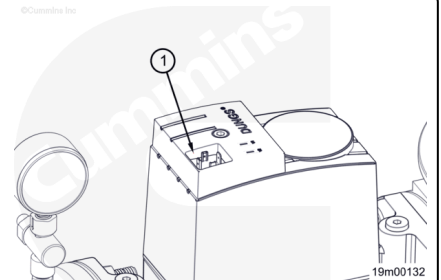


Make sure there is a rubber gasket and that there are no bent pins on the male fuel shutoff valve connector. Install a new gasket if necessary.

Align and connect C19 to the top of the fuel shutoff valve (1).

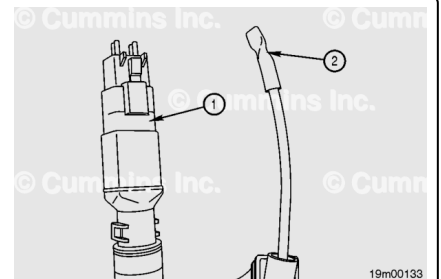
Install the captive screw and tighten.

Torque Value: 0.7 n•m [6 in-lb]

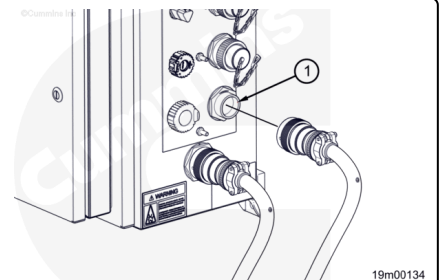


After the high-pressure switch, low-pressure switch, closure validation switch, and fuel shutoff valve have been connected, there will be two wire harness branches left for installation. The 4-pin Deutsch™ connector C30 (1) allows for the use of either a gas indication light or switch shutdown function. Consult the OEM and/or installing Cummins® Distributor for proper connector use and installation guidelines. If the connector is **not** used, securely tie-wrap the harness branch so that it does **not** affect Dual Fuel operation.

The bare harness stub (2) is currently **not** used in this system installation. Securely tie-wrap the harness branch so that it does **not** affect Dual Fuel operation.



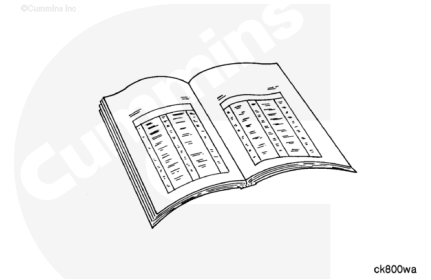
Connect the gas train wiring harness connector C4 to the Dual Fuel panel port C3 (1).



DOC Temperature Sensors

Secure the DOC temp sensor wiring harness to the DOC support frame. It is recommend to secure the harness directly above and below the connector.

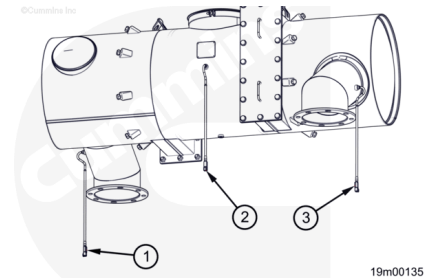
Note : No portion of the sensor cable or attached harness may come within 2.54 cm [1 in] of the aftertreatment body.



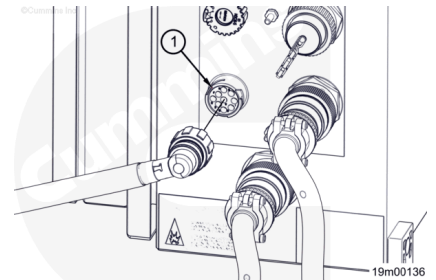
Connect connector C24-4 to the left DOC inlet temperature sensor (1).

Connect connector C24-3 to the right DOC inlet temperature sensor (2).

Connect connector C24-1 to the DOC outlet temperature sensor (3).



Connect the DOC wiring harness connector C12 to the Dual Fuel panel port C11 (1).

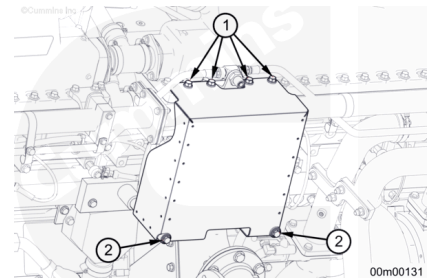


Debris Shield

Install the debris shield.

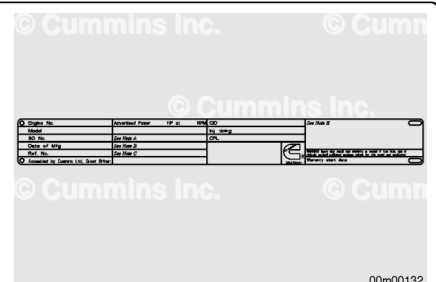
Tighten the six bolts - four short across the top, Cummins® Part Number 3069120 (1), two long on the bottom, Cummins® Part Number 137872 (2).

Torque Value: 38 n·m [28 ft-lb]



Emissions Dataplate Sticker

The Dual Fuel emissions dataplate sticker backing **must** be read and understood. After the dataplate sticker backing has been read, choose the correct dataplate sticker to place on top of the "Important Engine Information" section. This will either replace the T.P.E.M. or Environmental Protection Agency Tier 2 language bulletin. After all components have been installed, an emissions dataplate sticker will be placed on top of the existing "Important Engine Information" section of the QSK50



engine dataplate. This sticker will provide the pertinent engine fueling information to verify Dual Fuel engine operation meets or exceeds Tier 2 emissions guidelines.

Document History

Date	Details
2014-11-7	Module Created
2015-3-23	broken link
2015-7-6	Correct graphic and callout list
2015-7-16	fix link

Last Modified: 20-Jul-2015